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Workpackage J

Innovative Tourism Statistics

Deliverable J2: Interim technical report showing the preliminary results and a general description of the methods used.

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Prepared by:

WPJ Team

Workpackage Leader:

Marek Cierpiał-Wolan (Statistics Poland, Poland)
M.Cierpial-Wolan@stat.gov.pl
telephone : + 48 17 85 35 210 ext. 311
mobile phone : + 48 515 037 640

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Introduction

Over several years changes in tourism trends on the demand- and supply-sides have occurred. These changes have forced national statistical institutes (NSIs) to implement continuously innovative means for the verification of new sources of information, as well as, for the improvement of methods and techniques of obtaining data. An important element towards obtaining reliable and useful statistical measures are modern ways of combining statistical data with external data from various sources.

The development of creative, interactive and scalable methods of data integration is not a trivial problem. This is due, inter alia, to the different data structure, varying frequency and associated methodological differences for collecting and compiling data. In fact, developing innovative approaches to combine data from different sources is a challenge for today's official statistics.

The work undertaken in the ESSnet Big Data II: Workpackage J (Innovative Tourism Statistics) project is an attempt to meet this challenge, namely:

- make an inventory of data sources related to tourism statistics,
- identify, describe and classify source characteristics,
- develop scalable solutions/methods for data collection from internet using web scraping techniques,
- implement data linkage and matching techniques to integrate data sources,
- execute spatial-temporal disaggregation of the capacity of tourist accommodation establishments,
- combine data for expenditures estimation in country breakdown.

The project results will contribute to the development of an innovative (pilot) data integration system to improve the monitoring tasks of NSIs on tourism statistics.

This document consists of the following Chapters:

Chapter 1 describes the tasks carried out as part of the WPJ in the reported period and provides a brief summary of them.

Chapter 2 is devoted to the inventory of big data sources related to tourism statistics. In particular:

- Subchapter 2.1. contains analyses of identified data sources in the field of tourism;
- Subchapter 2.2. includes information on current works of partner countries on obtaining data from external data administrators;
- Subchapter 2.3. characterizes the Flow Model showing directions of combining and using various sources of data in tourism statistics proposed by the WPJ.

Chapters 3 relates to source characteristics and provides an example of an interactive graphical presentation of the relationships and hierarchies between variables and domains from big data and administrative sources and the corresponding ones from official statistics.

Chapter 4 is devoted to description of the solution used in web scraping. Innovative tool for retrieving data from unstructured sources is presented in Subchapter 4.1., while universal tool architecture in Subchapter 4.2. However, new developed by the WPJ tool for downloading data - Visual Modeler is shown in Subchapter 4.3. Issues encountered are discussed in Subchapter 4.4.

In Chapter 5 preliminary results of the linkage of data from web scraping with data from administrative registers and statistical survey frames are presented. Two approaches to data linkage are shown.

In subchapter 5.2. linkage of data from variety of sources based on addresses of tourist accommodation establishments was described and evaluated, while in subchapter 5.3. distance-based linkage of data was discussed and tested.

Chapter 6 includes the presentation of the method concerning spatial-temporal disaggregation of tourist accommodation establishments in terms of capacity.

The method of combining data for expenditures estimation in country breakdown is described in Chapter 7 including general description of the methods (Subchapter 7.1.) and preliminary results (Subchapter 7.2.).

Conclusions on the work carried out are provided in Chapter 8.

1. Characteristics of tasks performed under WPJ

The workpackage J in the ESSnet Big Data II project is implemented by eight European statistical offices: Statistics Poland represented by the Statistical Office in Rzeszów acting as the leader of the WPJ, National Statistical Institute of the Republic of Bulgaria (BNSI), Hellenic Statistical Authority (ELSTAT, Greece), Hesse Statistical Office (HESSE, Germany), Italian National Institute of Statistics (ISTAT), Statistics Netherlands (CBS), Statistics Portugal (INE) and Statistical Office of the Slovak Republic (SOSR).

In the period from 01/01/2019 to 30/09/2019 the partners completed a number of tasks. In the first stage, an inventory of external data sources and a selection of web scraped portals was carried out. Each of the above-mentioned partners reviewed national sources of information, selecting those from which data may be useful to improve official tourism statistics. During the inventory process, some project participants established a cooperation with data administrators in order to gain access to data that were previously unexploited or inaccessible to official statistics. All sources inventoried by the WPJ were described and catalogued taking into account their usefulness for estimating the demand- and supply-sides of tourism.

In order to characterise the sources in terms of identification, taxonomy, mapping and ontology, the methods used in this area have been reviewed and examples of R-language solutions have been proposed.

Based on external data sources WPJ partners developed the Flow Model concept describing the ways and directions of combining data from various sources.

At the same time, portals used for booking accommodation establishments were reviewed, broken down to those with global and local reach. It was decided that web scraping would include Hotels.com and Booking.com on a global scale. Description of the criteria for selecting portals, assumptions related to the implemented web scraping process and a preliminary outline of the process of creating a universal data acquisition method were described in the document *J1 - ESSnet Methods for webscraping, data processing and analyses*, which was developed by the WPJ at the end of March 2019. The web scraping method was constantly improved by eliminating the errors that were most frequently encountered by project partners when downloading data from the web. The methods of solving them were also characterized and described.

On the basis of experience gained during web scraping, a technical description of open source libraries and the technologies used were applied to create a universal tool that facilitates data downloading from unstructured sources. This tool will be used for web scraping of data in all partner countries. Technical information related to this tool are described in this document.

Using various methods, an attempt was made to unambiguously identify accommodation establishments (objects) that were found in web scraping databases, administrative registers and surveys for tourist accommodation establishments. Pairing of objects was performed by comparing address data, as well as using a more advanced approach, namely a solution based on the use of geographical coordinates of the location of objects.

Within the framework of the conducted works, the preliminary methodology of spatial temporal disaggregation of data from various sources was also developed. It includes, among others, adjustment of the types of objects on the Internet portal to the classification functioning in official

statistics, estimation of the number of beds. As a result, we will obtain higher frequency data broken down into smaller territorial units.

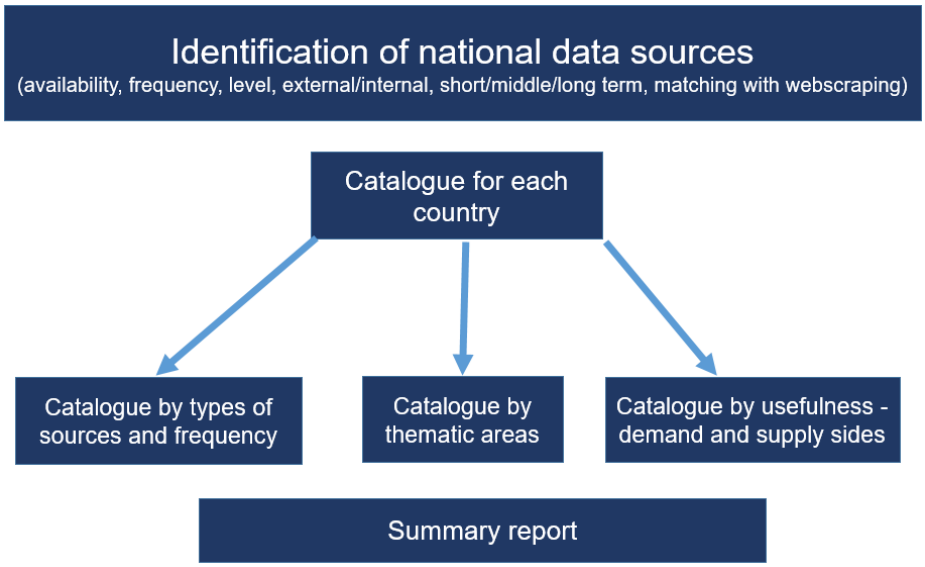
An important issue which was taken up within the framework of the works was an attempt to improve the quality of data on residents' expenditures. The method of estimating trips in air traffic developed within the ESSnet Big Data I project allowed the identification of countries to which residents travelled and which have not appeared in the sample survey so far. The continuation of this method in the current edition of the project is an attempt to estimate the expenses of residents travelling to newly identified countries. To this end, various methods were tested, including Machine Learning methods.

2. Inventory of big data sources related to tourism statistics

2.1. Identified data sources for tourism statistics

Data from external (administrative data, big data, etc.) sources and internal (statistical) sources of information will be one of the components of the created system. Inventory of sources was necessary to create a catalogue and determine the suitability of individual sources. During the work process, sources that can be used to improve the quality of official statistics in the field of tourism were selected. This process resulted in creating detailed catalogues that contained the characteristics of the identified sources. A simplified diagram of the inventory process is shown in Figure 1 below.

Figure 1. Diagram of the inventory process of data sources



As part of ongoing work, project partners have inventoried a total of 130 sources of information. Just over half of them (57.7%) were external sources, while the remaining 42.3% were internal sources. Considering the number of identified sources in individual countries, it was found that the Netherlands (44 sources), Slovakia (20), Poland (16) and Italy (14) showed the highest number of sources. The Netherlands showed the largest number of internal and external data sources (22 sources each). Among the remaining countries, Italy (9) and Slovakia (8) had the highest number of internal sources, while Poland (13) and Slovakia (12) collected the highest number of external sources. A detailed summary of the number of identified sources broken down by type is presented in Table 1.

Table 1. Number of identified data sources by type of source

Specification	Partner country							
	Bulgaria	Germany (HESSE)	Greece	Italy	the Nether-lands	Poland	Portugal	Slovakia
	Number of data sources [pcs.]							
Total	**6**	**12**	**11**	**14**	**44**	**16**	**7**	**20**
Internal sources	4	6	2	9	22	3	1	8
External sources	2	6	9	5	22	13	6	12

During the inventory, the identified sources were divided according to their usefulness in estimating the demand and supply side of tourism. Considering that some data sources can be used to estimate both sides of tourism at the same time, this division has also been included in the compilation (Table 2). As a result of the conducted works, it was found that among all the inventoried sources, those which data can be used for estimating the demand side of tourism have largely prevailed (82 sources).To estimate both the demand and supply 26 sources may be helpful for calculating and 22 for the supply side. The largest number of sources that could improve statistics on the demand side of tourism was identified by the Netherlands (25 sources), Slovakia (13), Poland (10) and Italy (9). In the case of sources useful for estimating the supply side of tourism the Netherlands identified the largest number of sources (8). The Netherlands also showed the highest number of sources that can be used to estimate both the demand and supply sides of tourism – over 40% of this type of sources was identified by all partner countries.

Table 2. Usefulness of identified data sources for demand or supply side of tourism

Specification	Partner country							
	Bulgaria	Germany (HESSE)	Greece	Italy	the Nether-lands	Poland	Portugal	Slovakia
	Number of data sources [pcs.]							
Total	**6**	**12**	**11**	**14**	**44**	**16**	**7**	**20**
Supply side	1	1	1	3	8	2	2	4
Demand side	5	9	6	9	25	10	5	13
Both	-	2	4	2	11	4	-	3

External data sources include those that partner countries do not yet have access to as well as those with limited availability. Analysing the availability of external data sources in all partner countries it was found that 52% of them were accessible. The highest share of available external sources was observed in Poland (almost 85%). Also Portugal showed a similar rate of availability of sources (83%). It is worth noting that among 11 online sources identified by partner countries, 10 were reported by the Netherlands and 1 by Poland. Table 3 presents availability of identified external data sources broken down into available and unavailable ones.

Table 3. Availability of external data sources

Specification	Partner country							
	Bulgaria	Germany (HESSE)	Greece	Italy	the Nether-lands	Poland	Portugal	Slovakia
	External data sources [in %]							
Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Available	100.0	33.3	22.2	40.0	59.1	84.6	83.3	16.7
Not available (temporarily or permanently)	-	66.7	77.8	60.0	40.9	15.4	16.7	83.3

Among the available external data sources, those useful for estimating the demand side of tourism prevail (Table 4). In Bulgaria, Germany and Greece all available external data sources are useful for estimating demand side of tourism. The Netherlands reported only 31% of available external data sources as useful for demand side of tourism, but almost 54% of data sources identified by this country can be useful for both side of tourism.

Table 4. Usefulness of available external data sources for demand or supply side of tourism

Specification	Partner country							
	Bulgaria	Germany (HESSE)	Greece	Italy	the Nether-lands	Poland	Portugal	Slovakia
	Available external data sources [in %]							
Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Supply side	-	-	-	50.0	15.4	9.1	40.0	50.0
Demand side	100.0	100.0	100.0	50.0	30.8	63.6	60.0	50.0
Both	-	-	-	-	53.8	27.3	-	-

The inventory of data sources by WPJ partners are characterized by varying frequencies. Most of them have a monthly frequency. Sources which data have weekly or even daily frequencies have also been shown. They constitute a very important element in the construction of the system supporting official data in the field of tourism due to the obtained high frequency time series, which will facilitate analysis and allow more accurate capture of changes.

Despite the official completion of work related to task 1b resulting from the adopted schedule, it should be emphasized that the inventory work is still underway. Project partners are identifying potential sources of information, especially those that can be used to support tourism statistics at the local level. The results of the work regarding the identification of this type of sources and the possibilities of their use will be presented in the final report of this project.

2.2. Review of works on obtaining data from external data administrators

In several countries discussions are currently in progress with data administrators regarding access to the databases that are in their possession.

Access to identified data in the possession of external data administrators, which has not been used by public statistics so far, requires establishing cooperation with the data administrators, determining conditions of access, concluding contracts and conducting arrangements regarding the scope and frequency of data acquisition, which is often a long and time-consuming process.

Hesse Statistical Office (HESSE, Germany) is involved in discussions with entities which possess data on financial transactions in the tourism sector (see workpackage G of ESSnet Big Data II related to financial transaction data. Additionally, some statistical offices in Germany already have started cooperation (while some plan to do so) with mobile network operators to investigate anonymous data from mobile networks, e.g. for population statistics (see workpackage I of ESSnet Big Data II on mobile network data). In a pilot study, the statistical office of HESSE investigates the use of such data in the field of tourism statistics to capture tourism flows currently not covered by official tourism statistics (e.g. same day visits, overnight tourisms including small accommodation businesses).

Partners from **Statistics Netherlands** (CBS) are negotiated with two mobile network operators (connected to WPI: Mobile Networks Data), but these data are not considered as available in the short-term for WPJ, as agreements with one telecom enterprise are quite restrictive and the agreements with two other telecom enterprises are still being negotiated.

Hellenic Statistical Authority (ELSTAT) intends to negotiate with the Tax Authority in order to get access to the Register of Short-term Lease Properties and the Database of Short-term Leases. Concerning the big data sources, ELSTAT is going to negotiate with data administrators on the access to data concerning traffic sensors located on the borders and future negotiations are also planned with the national mobile network operators.

The Statistical Office of the Slovak Republic (SORS) is establishing the conditions for access to data with the Municipality Office of the Capital City Bratislava and currently is at the stage of signing the agreement. Data will be available in a structure of: date, name of accommodation establishment, company registration number, number of nights spent (of which foreigners) and number of arrivals (of which foreigners). Data will be useful for identifying accommodation establishments which are not listed in the accommodation establishment register of the SORS, including establishment with less than 10 bed places. Moreover, the SORS has set the cooperation activities with Bratislava Tourist Board about data availability from Bratislava Tourist Cards. Concerning the big data sources, SORS has negotiated with mobile network operator Slovak Telekom and the Skytoll Company, which is the operator of the toll system in Slovakia.

National Institute of Statistics in Italy (ISTAT) is currently investigating the possibility of obtaining big data concerning mobile phone data - agreements with mobile phone data providers are in progress, but the authorization of Italian Privacy Authority are waiting. ISTAT have already been made experimental statistics for measurement of traffic flows on the Italian road network based on traffic images data. Additionally, financial transactions data could become available from WPG, ESSnet on Big Data II.

Statistical Office in Rzeszów (GUS) is conducting advanced discussions with the Town Hall of Rzeszów in order to obtain data from the Smart City system (S.C.). The S.C. information system resources include, among others, data on the intensity of urban road traffic from city monitoring cameras, data

on the number of users of city parking meters, as well as data on the number of people using public transport. The exchange of information between the Statistical Office in Rzeszów and the Town Hall will be conducted on the basis of a mutual agreement, which may open the way to cooperation between the mentioned parties in the field of tourism.

The results of ongoing negotiations of all project partners with data administrators, as well as the resulting progress in the project, will also be described in the final report.

2.3. Flow Model proposed for tourism statistics

On the basis of the data sources inventory, the project partners have developed Flow Models for their countries according to the accepted scheme. Each model presents proposed directions for combining data collected from external sources and web scraping with data from official statistics. An important issue regarding the Flow Model is the block breakdown of data sources that can be used to compile statistics on the demand and supply side of tourism. One important element of the models developed by particular countries was to present areas and directions in which the results of combining data from inventoried sources (using statistical methods) will be used to improve official statistics.

Proposed areas and directions:

- Improving the completeness of tourist accommodation base
  The use of web scraping method and geographical coordinates will improve both the completeness of survey population and improve the quality of results for tourist accommodation establishments.
- Spatial disaggregation of data on tourist accommodation base
  Data obtained from non-statistical sources (administrative and big data) will allow to obtain reliable information about tourist accommodation establishments at the lowest levels of territorial division (villages, districts, housing estates, etc.).
- Flash estimates of the use of tourist accommodation base
  Flash estimates in the field of tourism will respond to the growing demand from stakeholders regarding the rapidly changing situation on the tourism market. Monthly data concerning tourist accommodation establishments will be delivered within T+1 period.
- Improving the quality of data on trips
  The new data sources and statistical methods used (e.g. machine learning) will allow for verification of estimations of the size of tourist traffic according to travel directions, means of transport, types of accommodation.
- Improving the quality of tourists expenses data
  The use of data on prices of accommodation, meals, transport, ticket prices for tourist attractions will allow verification of estimation of the amount of expenses connected with trips.
- Improvement of satellite accounts
  Thanks to improving the quality of data on trips and tourists expenses the verification of indicators of the direct contribution of tourism to total value added or total gross domestic product (GDP) will be made.
- City tourism
  The information obtained from Smart City systems can be used to estimate changes in the urban population at a given point in time, e.g. during urban cultural events such as concerts, sporting events, etc. Information on the number plates of arriving and departing vehicles will help in determining the place of origin of a vehicle (country and abroad) as well as in estimating

the number of people visiting the city. Based on the developed and implemented solutions, a use case will be created, which can serve as a guideline for other partner countries.

- Event related tourism,
  Based on new data sources it will be possible to estimate the number of people traveling in connection with participation in fairs, conferences, etc., and the degree of utilization of municipal, tourist and gastronomic infrastructure.
- Tourism potential,
  Detailed and high-quality data on accommodation in combination with data on communication infrastructure and resources of the geographical environment will allow to present more complete picture of the tourist potential of regions, while better estimates of the number of tourists and expenses incurred by them will allow for a more precise determination of the utilization of tourism potential.
- Tourism attractiveness.
  Data obtained from various sources at low levels of territorial division will provide information on the basis of which it will be possible to develop indicators determining the tourist attractiveness of regions/cities.

The models developed by the project partners have been adapted to the number and types of inventoried sources in each country and to the areas where they will be applied in the short term. They are included in Annex 1 to this document, while their brief characteristics, taking into account the type of external data sources planned to be used and statistics potentially to be improved, broken down by country, is presented below.

The Netherlands to improve the quality of tourism supply side data, in addition to data from web scraping, intends to use information from the Register of non-categorized tourist accommodation establishments (Chamber of Commerce (via Business Register), Register of Addressen and Buildings (BAG 2019), and others (Locatus, Airports register). Among the external data sources that may be useful in estimating the demand side are, among others, data from research company on holiday, recreation (NBTC-NIPO) and other sources (i.e. Dutch Terrace Research - inventory of the prices across restaurants, pubs and bars of a standard set of beverages in the most well-known Dutch cities carried out by Van Spronsen & Partners, statistics on business travel, museums and recreation). The planned results include: improving the completeness of tourist accommodation base, spatial disaggregation of data on tourist accommodation base, improving the quality of tourist's expenditure data and improvement of tourism satellite accounts.

In **Poland**, the basis for improving the quality of data on tourist accommodation will be data obtained from web scraping. To improve quality, additional information will be used from the Register of non-categorized tourist accommodation establishments (EOT), which is run in Poland by commune offices. Information obtained from web scraping and data from Border guard data, Automatic Number Plate Recognition System ANPRS and Traffic sensors will be used to improve the quality of estimates of data on tourism demand. An important element of the demand side of tourism will be the data from the Smart City system (functioning in many cities), thanks to which it will be possible to determine the volume of tourist traffic at the local level. As a result of using statistical methods, i.e. combining and disaggregating data from these sources, it is planned to achieve several goals. These include: improving the completeness of the tourist accommodation establishments, improving the quality of data on tourists' trips and expenses, event tourism, developing flash estimates of the use of tourist accommodation establishments. The effects of the described work and the results obtained may be used in the future for improvement of Tourism Satellite Accounts (TSA).

In **Portugal**, estimates of the supply side of tourism will be compiled using data obtained from web scraping and data from National Tourism Registration - Tourist Establishments and National Tourism Registration - Local Accommodation. Data from tourism demand side surveys will be supported by data from web scraping and data from administrative sources - Airport Data, Credit and Debit Card Transactions. As a result of combining data from these sources, it is planned to achieve six expected results, including improvement of quality of tourists' expenses data, improvement of quality of data on trips, flash estimates for tourist accommodation and improvement of Tourism Satellite Accounts (TSA).

In **Italy**, the improvement of the quality of the supply side data could be achieved by using web scraping related data and data from administrative sources such as Water consumption and Waste production. The tourism demand side could be estimated based on web scraping data and using Survey on museums and similar institutions data, Water consumption data, Railway, airport and port data and Border Traffic survey (BI) data. On the basis of the data combination, ISTAT will aim to develop results that can contribute to the achievement of the 10 objectives. These include, among others, the following: Improving the quality of data on trips, improving the quality of tourists' expenses data, improving the completeness of tourist accommodation base, tourism potential and others.

Bulgaria has developed a Flow Model in which data from the Ministry of Interior, Airports and others will be used to improve the quality of tourism supply side data. In the case of the demand side of tourism, apart from the data obtained using the web scraping method, Bulgaria intends to obtain additional information from the same sources as for the supply side. Based on data from the abovementioned sources, Bulgaria intends to obtain results for improving the completeness of tourist accommodation base, flash estimates of the use of tourist accommodation base, improving the quality of data on trips, improvement of satellite accounts.

In the Flow Model developed by **Slovakia**, the basis for improving the quality of tourism supply data, in addition to data from web scraping, will be data from Regulatory reporting FIN1-12 and Monthly bed tax data including the overnights data in the cities Bratislava and Košice. The tourism demand side in Slovakia will be additionally estimated based on data from web scraping and from an external source: Financial transaction data. As a result of using WPJ methods of combining data, Slovakia will attempt to improve the quality of data on tourism trips and improve the completeness and data validation of tourist accommodation establishments. Improving the quality of data in this area will allow the Improvement of Tourism Satellite Accounts (TSA) in the future. In addition, data from BTB data (unique Wi-Fi connection, etc.) can be used to determine Tourism potential, Tourism in selected cities or regions, Tourism centre points (attractions).

Germany (HESSE) improves data quality of the supply side of tourism by using several data sources gathered by web scraping and linked to official statistics (accommodation survey and business register) in order to improve completeness of official tourism statistics as well as improve plausibility checks of survey data with data stemming from web scraping. Additionally, HESSE established a cooperation in order to use anonymous mobile network data to measure tourism flows not covered by official statistics (day visits, overnight stays in small accommodation businesses). This will contribute to improve the picture of the demand side of tourism. Other external data sources include the number of visitors and travellers by plane and ship which contribute to the demand side of tourism. Given data from financial transactions regarding the "sharing economy" become available from cooperation with data holder (see workpackage G), they could be used to complete the picture

on the demand side of tourism. Additionally, there are some surveys strongly related to tourism and expenses of tourists (but not conducted by the HESSE). Either they have a very strong regional focus or they do not contain information on expenses.

Greece has developed a model which focused on improving the quality of tourism supply side data. Estimates will be developed using data from web scraping and external data sources such as the Tax Authority Register of Short-Term Lease Properties and Register of properties in Greece, which are offered for short-term lease through digital platforms. As a result of merging the data, Greece intends to achieve the following goals: improving the completeness of tourist accommodation base, flash estimates of the use of tourist accommodation base, spatial disaggregation of data on tourist accommodation base.

Currently, all partner countries have been scraping data from Internet portals used for booking accommodation establishments (Hotels.com, Booking.com).

Scrapped are data on:

- offer Id,
- hotel ID,
- hotel name,
- hotel address,
- type of accommodation,
- star rating,
- number of guest reviews.

These data will be used to improve the quality of statistics on the supply side of tourism.

In subsequent stages of work, web scraping will be used to obtain data from portals containing also information on tourist trips and tourist expenses.

From Skyscanner.net it is planned to extract the following data:

- date of departures,
- departure airport,
- destination country,
- destination airport,
- transfer airport,
- name of the carrier,
- flight number.

From Booking.com the data on airplanes and local transport will be scrapped:

- on airplanes:
  - flight number,
  - type of aircraft
- on local transport:
  - car rental price,
  - brand of a car,
  - duration of trip,
  - price of taxis from/to the airport.

From seatguru.com data on type of aircraft and number of seats in the plane will be gathered.

These data with the use of mathematical methods and models, will serve to improve the quality of the demand side of tourism.

3. Source characteristics

In the objectives defined for WPJ – Innovative Tourism Statistics, it is noted that "The increasing growth of information leads to big data and information systems targeting to administer, analyse, aggregate and visualise these data". Variable or data issues, such as overlapping concepts, indirect relations between sources and redundancies of information, although not always obvious or immediately recognizable, are inevitable when combining big data sources with administrative sources and official statistics.

Big data as well as administrative data are not produced for the purpose of official statistics and, therefore, before integrating them in the production system of official statistics, potential quality, methodological, legal, privacy and other issues should be identified and addressed while, when multiple sources are to be combined their indirect interconnections should also be taken into account.

As the information expands horizontally (more domains or areas) and vertically (more information for each domain or area), it will be increasingly difficult to manage, search and extract implicit information. Therefore, it is strategically significant to prepare and to present the relevant information with an efficient interface.

Task 1c - Inventory of big data sources related to tourism statistics: source characteristics - has four aspects:

- Identification: to answer the question which variables are relevant?
- Taxonomy: to enable the classification of variables into topics i.e. where each variable belongs to?
- Mapping: to establish direct and/or indirect relationships between variables from data sources coming often from non-statistical offices to those from national statistical offices (NSIs).
- Ontology: to describe a subset of core variables, their (inter)relationships (eventually proxies) and hierarchies.

These aspects are closely related to the work developed in ESSnet Big Data I, WP-8 Quality, in particular the Deliverable 8.2 *Report describing the quality aspects of Big Data for official Statistics* and the work being currently developed in ESSnet Big Data II, WP-K Methodology and Quality. Therefore, this task is expected to integrate and relate to previous work done as well as work being currently done in the frame of ESSnet Big Data Projects.

3.1 Source and variable identification

This aspect of the task 1c concerns the most detailed and disaggregated level of information that is required for both sources and variables. A critical issue is to define the necessary level of detail as opposed to the most detailed one. The work begins by defining attributes and metadata but it may be expanded to include also, for example, the type of use, such as:

- calibration / validation of statistical indicators,
- partial / total complement or replacement of currently used sources,
- complement sampling reference frame.

An alternative is to use pragmatic approaches such as the use of *broader*, *specific* and *related* terms and eventually *non-preferred* terms (see Annex 2).

3.2 Source and variable taxonomy

With information based on source and variable identification (attributes, metadata, etc.) detailed at the appropriate level, it will be possible to explore the benefits of cluster analysis for an empirical based taxonomy. Once validity of clustering is properly assessed, the resulting clusters could be used as taxonomy classes.

Another possible approach is to explore *Big Data Typification Example for mobile network operators (MNOs)*, which is being developed by WPK. In this example the big data source is described according to three aspects:

- Source and Access: data ownership, legal, ethical and technical aspects, for example
- Metadata: this aspect has a very wide range; it can be the unit of measurement (base unit) or to a wider extend the variable quality issues.
- Data Type: Structured or unstructured data? Human or machine produced data?

For more in depth information on this typification, please consult WPK's documentation[1].

3.3 Source and variable mapping and ontology

This aspect of task 1c will address the source and variable mapping concerning official statistics variables and domains. The goal is to provide an interactive graphical representation of the relationships and hierarchies between variables and domains from big data and administrative sources and the corresponding ones from official statistics.

This approach is inspired by network analysis, which can be considered as a map of relationships. These relationships are composed by nodes (for example, statistical domains or data sources) and edges that represent the connections between nodes. In this map of relationships not only different types of nodes but also different types or intensities of connections (edges) can represented. For example, if a particular source is very reliable or its data is highly relevant to a particular domain, a thicker line than otherwise can graphically represent its connection.

Source and variable identification and taxonomy are part of a bottom-up process to produce a basis for an ontology. As sources and variables are identified in detail it will be possible to organize them in classes with an underlying hierarchy structure. The end result will provide domain knowledge within a shared framework and contribute to reduce ambiguities and misinterpretations.

3.4 R Packages for network visualisation

Although dedicated commercial and open source[2] software for ontology development is available, it is possible to cover the basic principles with an already familiar language such as R. This language offers some advantages as it is an already known tool in NSIs (or becoming more and more used also within NSIs). As such, it will be probably not only faster to produce results but also to integrate them in current existing workflows and to share data and code. Discarding the need to learn a new tool is also an important issue to consider.

[1] https://webgate.ec.europa.eu/fpfis/wikis/display/EstatBigData/Big+Data+Typification.
[2] https://protege.stanford.edu/

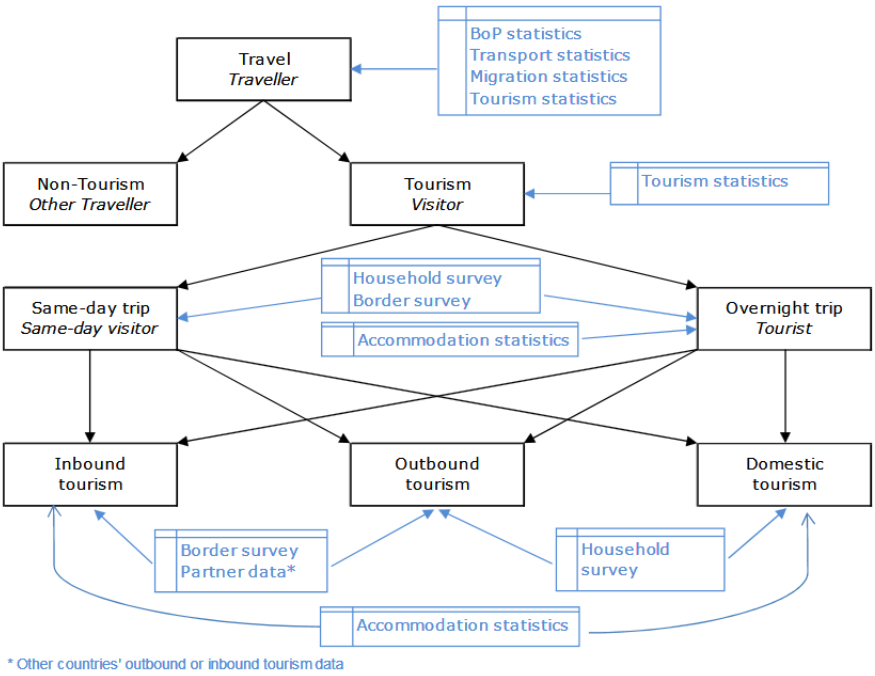
There are two R packages that are useful for this task: networkD3[3] and visNetwork[4]. With networkD3 it is possible to create D3 JavaScript Network Graphs from R such as 'JavaScript' network, tree, dendogram, and Sankey graphs. On the other hand, visNetwork provides an R interface to the 'vis.js' JavaScript charting library and allows an interactive visualization of networks.

They are both based on HTML widgets, so they are compatible with shiny, R Markdown documents, and RStudio viewer. R's ability to read and write in a multitude of formats makes it very appealing to this sort of task. In future work, if it becomes necessary to adopt a new tool, all the work produced could very likely be used.

3.5 An example: Scheme for Travel and Tourism (EUROSTAT)

To test the feasibility of this solution, we chose a well-known example from tourism official statistics. In EUROSTAT's *Methodological Manual for Tourism Statistics 2014* v.3.1 a basic scheme that represents flows (connection and direction) between possible sources and tourism flows can be found (Figure 2).

Figure 2. Scheme for Travel and Tourism (EUROSTAT)



Source: EUROSTAT - Methodological manual for tourism statistics — 2014, v. 3.1 (pp18).

This is a good example of what needs to be produced in task 1c because it represents hierarchies, domains, concepts and its relations. The layout of this static scheme is perfectly adequate for a limited amount of information to be displayed. But if the user needs to add more sources, variables and its relations, it would become cluttered and very difficult to read, interpret or even find relevant

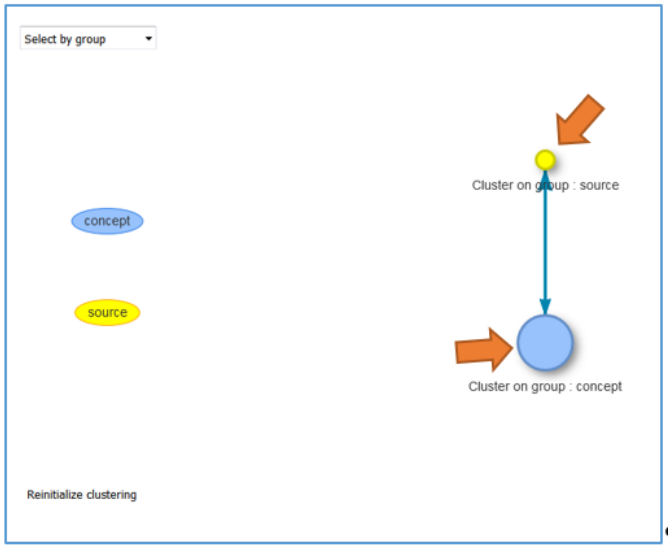
[3] https://christophergandrud.github.io/networkD3/
[4] https://datastorm-open.github.io/visNetwork/

information. Similar problem occurs with WPJ as there is a significant number of sources already identified.

The code[5] in Annex 3 replicates the data and the scheme but in an interactive mode. This is done by using the R packages visNetwork and networkD3 and exporting the resulting object to an HTML file that can be opened with a common web browser (in this case Google Chrome was used).

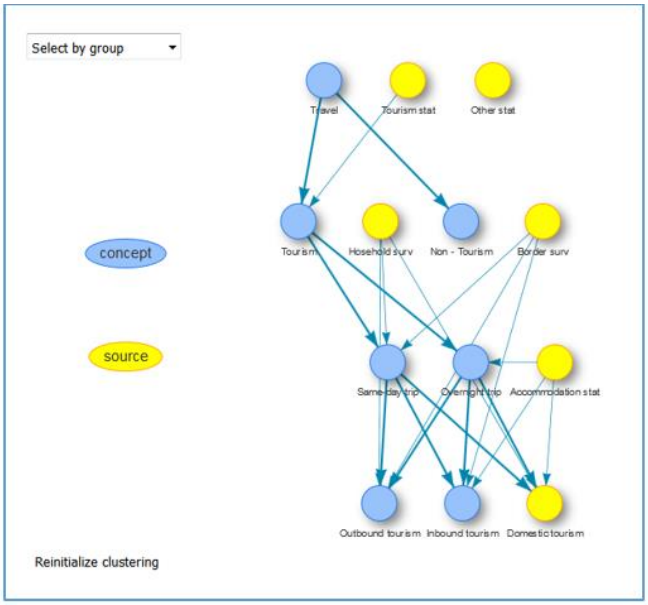
When opened in a web browser, the visNetTest.html file shows the network in a compressed mode.

Figure 3. Step one: compressed network



By clicking the nodes of both groups (concept and source), the network expands and shows all its components. The layout is automatically produced according the input parameters.

Figure 4. Step two: fully expanded network

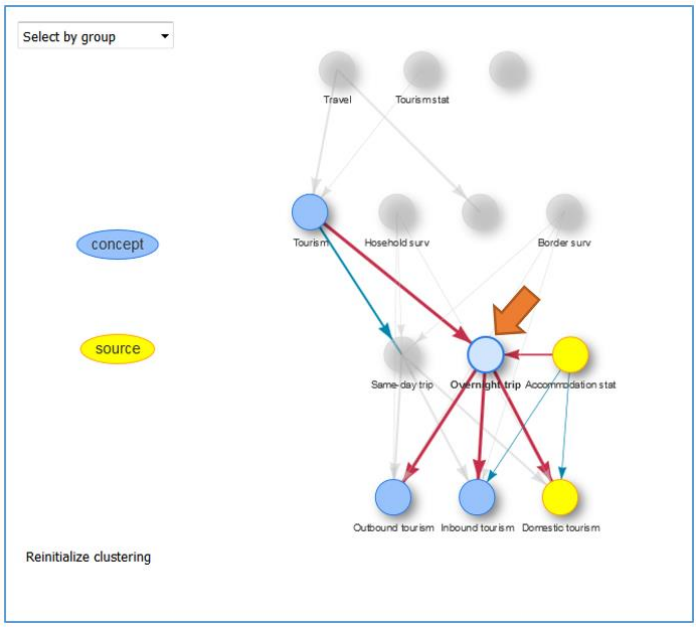


[5] This code used R version 3.6.1 (2019-07-05), RStudio version 1.2.1335 and packages: dplyr 0.8.3, visNetwork 2.0.8 and networkD3 0.4.

The hierarchies are automatically represented based on the information described in the data (edges: from -> to). The nodes can be dragged from side to side to facilitate the visualisation but the connections themselves are never broken.

Another interesting feature is that by clicking a node, for example *overnight trip* (Figure 4), the direct relationships being either concepts or sources can be isolated. In a more complex network, the extent of this relation can be defined to show up to the 2nd, 3rd or nth level, for example. This is particularly useful in a heavily dense network.

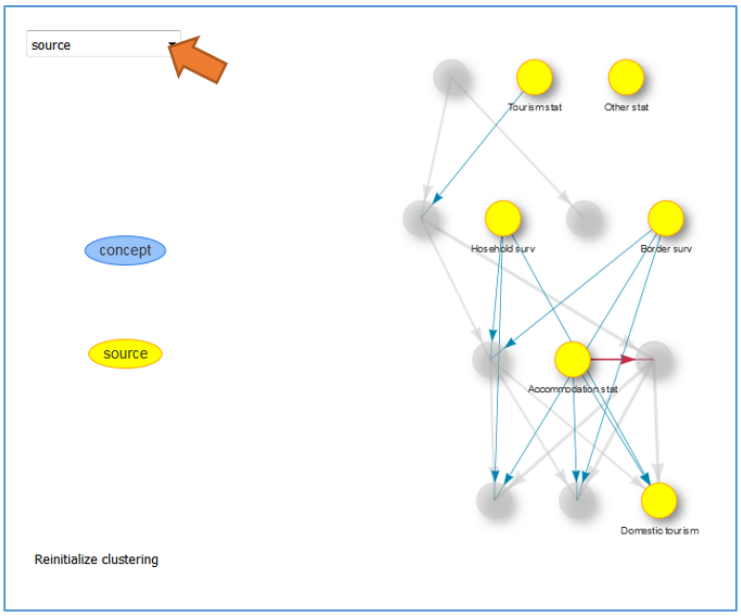
Figure 5. Step three: isolate direct relation to a node



The visNetwork objects has highly customizable basic parameters such as shapes, colours, fonts but also more sophisticated ones like layouts, physics and interaction.

On the top left corner there is a combo box where one can select either sources or concepts allowing a more clear view on the interconnections of the represented sources.

Figure 6. Step four: use combo-box to select sources only



Further development can be implemented in order to display more detailed information on each represented node. The code in Annex 4 illustrates how it is possible to open an external HTML page upon a click on a node. This HTML page can be exported from a R Markdown report and contain text, tables, graphs and links, for example. To a certain extent it can be considered as a dynamic dashboard. This way it is possible to display very detailed information about sources, domains, concept or other kind of information represented by network nodes without the inevitable cluttering. The user can interactively explore the data by collapsing or expanding nodes and read more in depth level of information just by clicking a node which will open a new HTML page.

4. Web scraping solution

4.1 Innovative universal tool

The application of the web scraping method based on remote extraction of indicated information from websites is an effective but time-consuming task. It results, among other things, from the high dynamics of changes that occur on web portals, and thus also from the involvement of qualified programmers. It is also related to the use of various technologies by Internet portal. They are characterized by a diversified structure, which significantly hinders the rapid acquisition of data, their cleansing and aggregation. On the basis of observations and its own experience, WPJ started to work on the creation of a universal and automated tool for retrieving data from unstructured sources. The proposed solution will enable a person with no specialist knowledge to create and launch a scraping process for any portal. The created system will allow one to create a scraper with a simple to use interface. Additionally, the developed method of downloading data will imitate the user's behaviour in the network. Thanks to this, it will avoid excessive load on web scraped portals.

Depending on the source of information we use, data can be obtained passively or actively with the help of web scraping. In the **passive** mode, the data are taken from simple (static) web pages, which do not contain mechanisms that require interaction with the user. Data downloaded in the passive mode are then saved in the form of HTML, XML or TXT files. Currently, fewer and fewer websites allow the use of passive mode. This is due to the structure of most of the available websites, which are characterized by a much higher visual attractiveness, and thus a more modern design. The use of the passive method to download data from this type of sites is therefore very limited or in many cases impossible to implement. In the **active** (dynamic) mode, the content on the website is generated in an asynchronous way. Depending on the user's interaction with the portal, the data visible on the screen are automatically updated. The development of technologies used in web browsers to present content (including JavaScript, Ajax, HTML5, WebGL) allows one to browse sites with a rich user interface, as well as the use of advanced options for filtering and searching for data. Web scraping conducted on such portals using the above technologies with the use of simple tools does not allow to collect complete data. In view of the need to retrieve data from modern portals and the potential development of presentation techniques, the following guidelines have been developed by WPJ, which should be followed in order to create a universal data collection tool:

- using the service in the passive and active mode,
- the possibility of conducting interactions on the portal according to an established algorithm of conduct simulating the user's behaviour,
- saving the status of the page with all the information on it, for further processing in OFFLINE mode (usually in the form of HTML files),

- conversion to UTF-8 pages containing different character encoding, due to the need to search for and combine characters, words and phrase.

Due to the difficulties encountered during web scraping of websites, based on the above guidelines, an attempt was made to create an effective data retrieving tool based on an innovative query algorithm. The work started with the analysis of the most popular available solutions, tools and libraries for retrieving data from portals, including:

- Scraper API,
- Smartproxy,
- Octoparse,
- ParseHub,
- Scrapy,
- Diffbot,
- Cheerio,
- HTMLCleaner,
- JSoup,
- Beautiful Soup,
- Puppeteer.

Due to the need to retrieve data from portals that require continuous user interaction, only Puppeteer meets certain guidelines. However, Puppeteer does not implement the mechanisms of interaction scenario building, HTML data processing (e.g. in case of dynamically generated links to subpages), and downloading data in the form of a uniform structure for different language versions of the portal. Other tools allow you to download the code of static pages in text or HTML form, and also provide simple navigation with the use of static links to subpages.

4.2 Universal tool architecture

Because none of the analysed solutions allows for downloading complete data, a universal solution was created based on:

- Java,
- WebKit - Chromium,
- Java communication library - Chromium (Selenium),
- HTML parser (own Java library),
- graphical environment for WYSIWYG interaction modelling, including its own library of visual components, suitable for designing logic, I/O operations, parsing data, saving to a file.

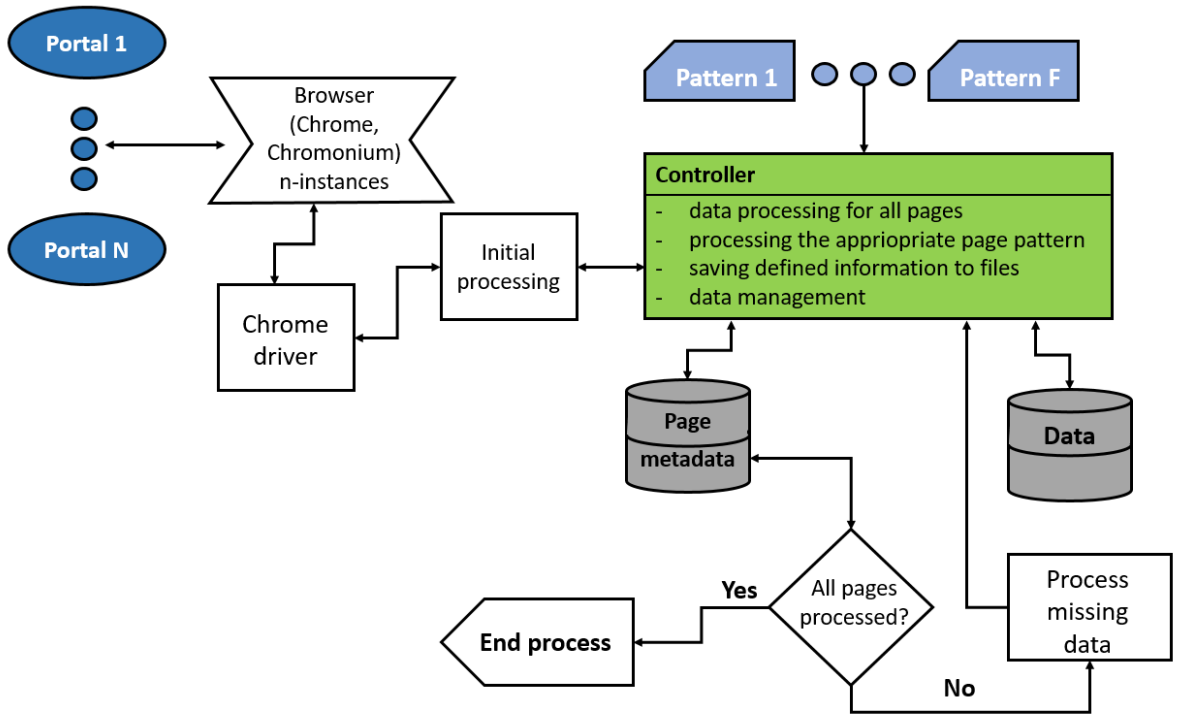
The following technologies were used to build the system:

- Java 8 64-bit (version 1.8.0.60),
- chromedriver_2_44 (Windows, Linux), or later,
- chrome_32_64_0_3282_140, Brave 0.68.140,
- selenium-java-3.6.0,
- NetBeans 8.02 (development environment).

The assumptions made and the technical solutions applied at the testing stage met all the requirements for building a tool useful for downloading data from portals offering accommodation

services. A diagram of the architecture of the created system taking into account the above mentioned guidelines, applied technologies and worked out solutions are presented in the Figure 7.

Figure 7. Simplified diagram of the architecture tool

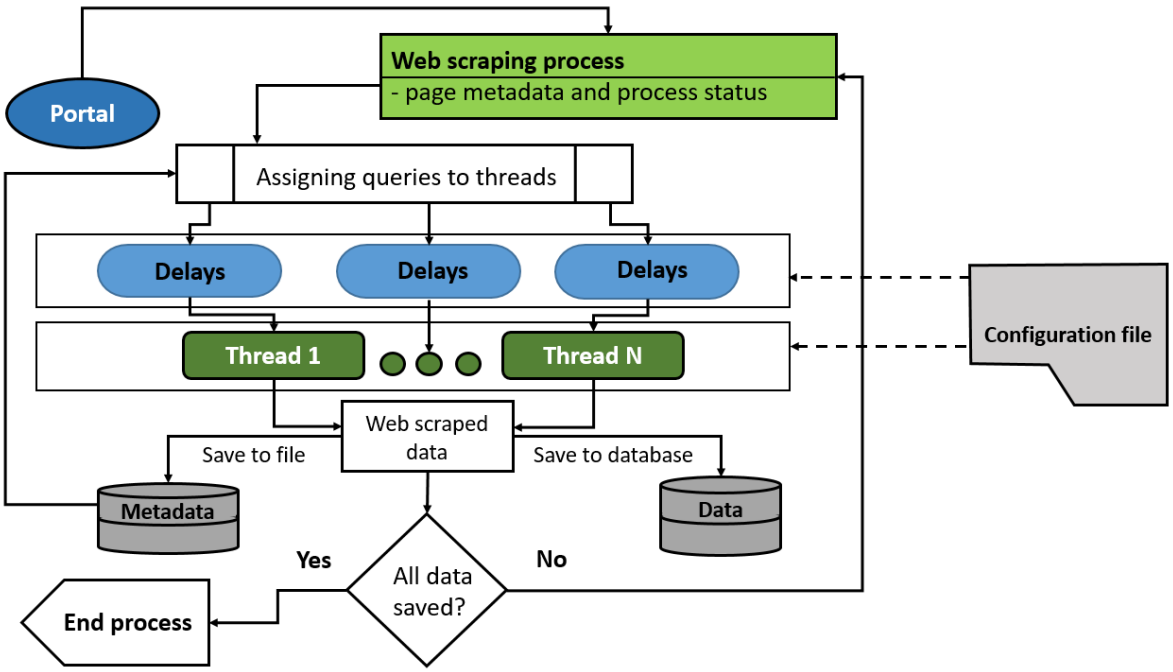


When using a web scraping method, there may be a number of unexpected problems that may affect data retrieval. Very often, information published on web pages ceases to be available to users over time. In practice, this prevents access to historical data, and such data is necessary for long-term analysis. The layout of the web portal is a common problem. Websites change their structure (page layout, location of its individual elements, etc.), which makes it impossible to develop a single way of interaction/parsing the portal without the need for programmers to interfere with the tool code. As a result, people modifying the code must constantly control the process without having detailed information on which stage of the process an error that prevents the downloading of data occurred. Thanks to the use of interaction patterns in the tool, it is possible not only to easily identify the stage of the process causing the error, but also to quickly make the necessary changes. Availability of web portals is also subject to changes. Often there are technical breaks and many other unforeseen situations that prevent access to content that is the object of our interest. In order to optimize the data retrieval process, an iterative algorithm of web scraping was used in the tool. In each iteration, the tool collects and saves data in a specific moment, in the form of HTML files and metadata of the completeness of the set. The iteration is repeated until the complete set of data is obtained.

According to the document created within the ESSnet Big Data II project by workpackage C *ESS-web-scraping policy template*, all created web scraping tools should not burden the portal from which the data is downloaded. For this reason, the algorithm should execute further queries to the server with time intervals. Failure to follow the guidelines proposed by WPC is connected with the risk of treating the queries by the portal as a DDoS-type attack, which is connected with blocking access to our IP address. Therefore, a mechanism of differentiating time intervals delaying sending requests to web

scraped web pages has been introduced into the created tool. Additionally, the developed solution allows one to define different intervals for each web scrapped service. The implementation of the application in Java and the use of additional libraries that can be used on different hardware platforms allows you to run the created tool both in Windows and Linux. The optimization works carried out during the development of the tool contributed to the achievement of high efficiency of the created solution regardless of the platform used. The issues described above were used to create a universal tool, whose process flow diagram is presented in the Figure 8 below.

Figure 8. Multi-threaded system implementation



4.3 Visual Modeler – proprietary tool for downloading data

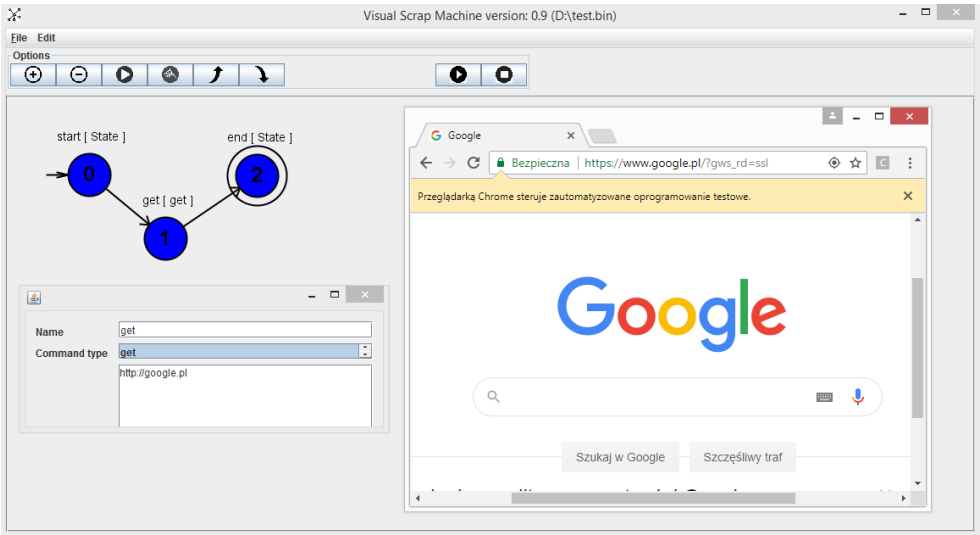
One of the key assumptions adopted during the design of the tool used to retrieve data from web portals was the ability to quickly and effectively analyse the entire process in the created solution. For this purpose, a graphical interface was created, which in an easy way allows one to initiate the process of web scraping for any portal not only by programmers, but also by people who do not have specialist IT knowledge. The classic method of web scraping requires a lot of work not only in terms of preparing the handling of the process for a selected portal, but also a great deal of knowledge of various libraries and built-in functionalities (Figure 9).

Figure 9. Part of code to retrieve data from the page

```
Source  History
443         try {
            ExpectedCondition<Boolean> expectation = new ExpectedCondition<Boolean>() {
445             @Override
                public Boolean apply(WebDriver f) {
447                 return ((JavascriptExecutor) driver).executeScript("return document.readyState").toString().equals(
448                 }
449             };
450             WebDriverWait wait = new WebDriverWait(driver, 30);
451             wait.until(expectation);
452
453             WebElement w_start = driver.findElement(By.cssSelector("a[class='xpb__link']"));
454             w_start.click();
455
456             Object[] windows = driver.getWindowHandles().toArray();
457             driver.switchTo().window((String) windows[windows.length - 1]);
458             Thread.sleep(3000);
459
460         } catch (Exception e) {
                e.printStackTrace();
462         }
```

In the tool created by WPJ, visual environments for modelling the web scraping process have been designed. Creating a web scraping process has been effectively facilitated. The process of its launch consists in assigning appropriate commands to individual nods and entering only key information needed to retrieve data from portals. The mechanisms implemented in the tool automatically process the information entered by the person creating the web scraping process into a set of instructions that perform specific tasks. The Figure 10 shows how to initialize the process described above and call the Hotels.com website.

Figure 10. Example of a diagram downloading a selected web page (corresponding code from Figure 9)



The main advantages of the developed visual environment for modelling the web scraping process include:

- saving the time needed to start the process,
- easy and quick possibility to make changes in the web scraping process of the portal,
- simplified analysis of the correctness of the created process and quick identification of errors,
- embedded functionalities to provide guidance as proposed by WPC,
- only basic knowledge of HTML and CSS is required,
- possibility to write/read processes,
- possibility to export data to files of different formats,

- easy extension of the functionality of the tool by adding additional modules,
- processes do not require compilation, they can be run directly after loading into the environment,
- possibility to work without modifications in Linux, Windows, Mac OS.

4.4 Issues encountered

The works consisting in the preparation of an IT structure necessary to launch the web scraping process of Hotels.com and Booking.com portals were initiated in January 2019 by a team from Poland. Depending on the capabilities and experience of other partners in conducting such activities, the method was introduced in the same time or with a slight delay. Currently, in each country, information from Internet portals is obtained on a daily basis. Downloaded data is saved in single files in MS Excel (XLSX) or CSV format. One file contains data about accommodation establishments and offers available at a given moment of launching the web scraping process. Depending on the popularity of the portals in the countries participating in the project, the one-time process lasts from 30 minutes to 1 hour. In the saved files, the following types of variables can be distinguished:

date - date of the scraping process,

scrapingType - variable describing defined parameters for web scraping,

destinationName - name of the region/city/country where the accommodation unit is located,

adults - number of adults renting accommodation,

children - number of children using accommodation,

offerId - ID of the offer on the portal,

hotelId - ID of the accommodation facility,

hotelName - name of the accommodation facility,

accType - type of accommodation facility,

locality - location of the accommodation facility (locality),

postalCode - address code of the place where the accommodation facility is located,

region - location of the accommodation facility by territorial division,

street - street at which the accommodation facility is located (address),

price - price of renting an accommodation facility,

starRating - accommodation facility standard,

guestReviewsRating - average rating of users resulting from their ratings,

guestReviewsTotal - total number of opinions issued by users,

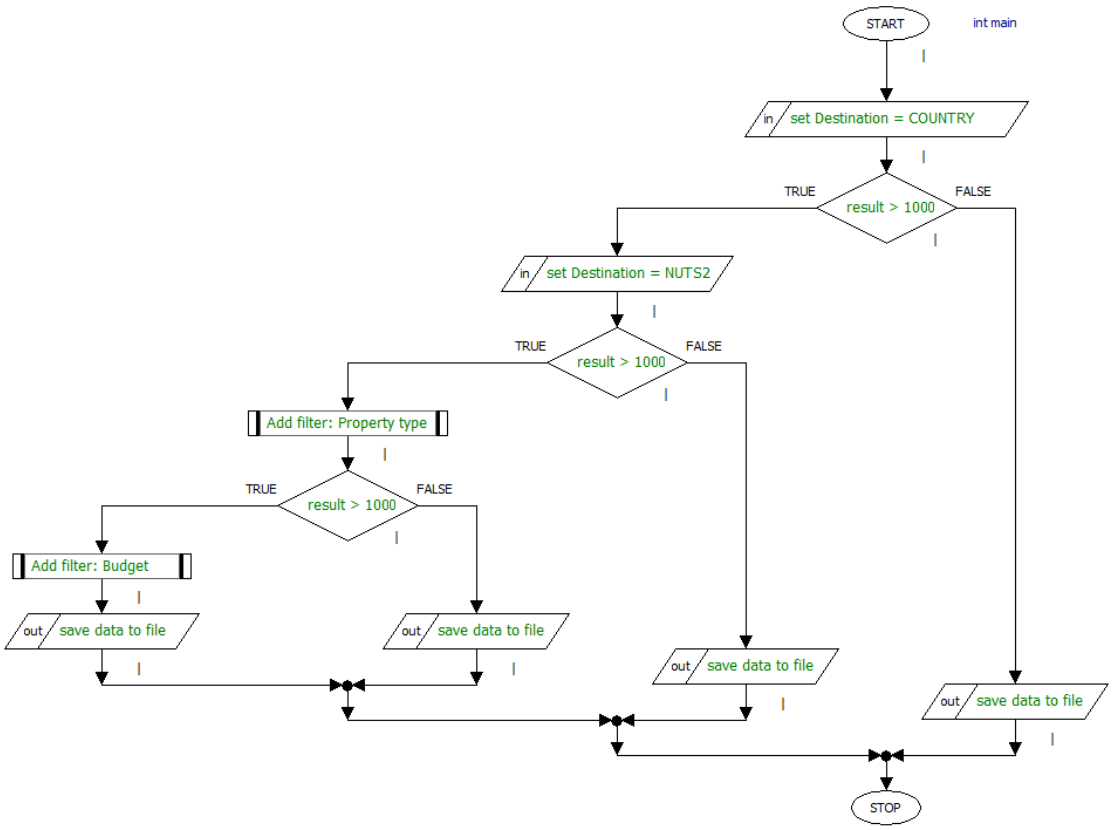
percentage - variable specifying the degree of use of objects for a given location.

In all countries involved in task 1a, a number of problems occurred during web scrapings, which required intervention of programmers. The most important problems are described below and along with the ways to solve them.

One of the main problems was the diversity of solutions used by portal developers, so it was not related to the preparation of the script from the technical side. For Booking.com - if the definition of the location is too general (e.g. Poland, Germany), the number of displayed offers is limited to 1000 by the creators of the website even if the number of offers found is much higher.

A solution may be to provide a more precise location (region, city) as a search criterion, in special cases to limit it to the type of accommodation establishments. If a country is selected as a location, only about 8% of the web scraping data is downloaded and saved in the database. When moving to NUTS2 territorial level, the number of results saved in the database increases to about 80%. Figure 11 is a web scraping diagram for Booking.com showing the process of maximizing the number of offers for the accommodation establishment saved in the database.

Figure 11. Process of maximizing the number of offers for the accommodation establishment saved in the database.



In order to carry out web scraping on Hotels.com, a script was created to initiate this process. The script was assigned unique identifiers of the location of accommodation establishments, which are part of their www address. Thanks to identifiers, web scraped accommodation establishments were identified on the basis of the assigned location. However, due to the changing identifiers of individual objects on the website, it was necessary to frequently update the script code. Updating the code and adapting it to the needs of all WPJ partners was a time-consuming task.

As a temporary solution, Portugal prepared a machine to create a table with locations for which data are to be collected. This significantly shortened the time of preparing the web scraping process, but it did not solve the problem of manual assigning an identifier. Ultimately, a solution was prepared in the form of a module that automatically searches for and assigns appropriate identifiers to the location. The new functionality solved all the problems with assigning identifiers to a location, which allows quick modification and adding new types of queries. The side effect of the proposed solution is an extended time of downloading data from portals.

In order for the web scraping process to work properly, all scripts must use browsers available on the market. Solutions created by WPJ have been optimized for the currently most popular browser Chrome. Unfortunately, frequent updates in particular browser versions and the range of changes made it necessary to constantly test and check how the tools behave after each update. An additional problem is the multitude of hardware configurations on which scripts are run. Few partners in the project, mainly Germany and Greece, encountered a problem with the tool for gathering data from the portal Booking.com. Starting new web scraping process stops all running instances of Chrome browser. If the instances are in use, they cannot be stopped and the script falls into a loop, initiating a driver used for web scraping infinitely.

The solution to this problem was introduced with new universal solution created by WPJ that use open source Brave browser built-in into universal tool. It gives creators of script's control when it comes to applying new versions as well.

The cases described were only part of the problems encountered during the work. However, they show the level of complexity of web scraping processes handling for particular portals and their diversity. Therefore, WPJ's work focused on the creation of a universal tool that will support all web scraping processes at the European level.

5. Data linkage – preliminary results

5.1. Introduction to data linkage

The linkage of data from web scraping with data from administrative registers and statistical survey frames is largely dependent on an appropriate matching strategy between these sources. Therefore, achieving a reliable and as complete as possible result in terms of the population of tourist accommodation establishments requires cross-checking the aggregates available on tourism-related Internet portals and the aggregates from the administrative records and surveys of tourist accommodation establishments.

When the data from portals are collected, the first question that arises pertains to the number of the establishments which occurred in the portal data as well as in the survey frame. It may be implemented by data linkage (or data matching) of data from the survey and data from accommodation portals. There are two main approaches to that task: deterministic (or rule-based) record linkage and probabilistic (or fuzzy) record linkage. In the case of the simplest deterministic matching, an identifier or a group of identifiers across databases are compared. Two records are linked whenever all of identifiers agree. One can rely on a single unique identifier, e.g. ID in Business Register. If it is not available then a group of identifiers may be used e.g. postal code and address. It may turn out that the combination of identifiers is not unique.

When the datasets come in different shapes and quality, creating scenarios must be taken into consideration in building a linkage algorithm. Postal codes and addresses in online portal, e.g. accommodation portal, often contain typographical and data entry errors, such as transposed digits and misspellings. That situation may occur if the portal does not force a standard format when entering data.

5.2. Address-based linkage

In the first attempt datasets were linked by addresses. Various formats of the postal codes and addresses were encountered in the data from the portal. Table 5 presents several examples of valid and invalid postal code with the format: two digits, dash, three digits.

Table 5. Valid and invalid postal codes

Valid	Invalid	Comment
35-959	35 959	There is space instead of dash
01-119	1-119	Zero at the beginning is omitted
24-003	pl 24 003	Country identifier and space are added
35-030	35030	There is no dash

If there is no standard format then it is not possible to link databases properly through deterministic linkage. In such a case joint database will contain false duplicates. Therefore, postal codes and addresses need to be standardized, which means they need to be brought to the same format. Many rules standardizing text strings may be implemented with the so-called regular expressions. Standardization is recommended for every type of data linking. For instance, replacement of space with dash in the postal code can be done with R code in the following way:

```
Ind=grep(pattern ='^\\d\\d\\s\\d\\d\\d\\s$',postalCode) #removing space at the end
postalCode[Ind]= gsub(pattern ='\\s' ,replacement ="", postalCode),
```

where \\d stands for digit, \\s is a space, ^ and $ marks the beginning and the end of a string, respectively. The first line finds records such as 35 959 and the latter one fixes it.
Unfortunately, such standardization is difficult for addresses. Variety of names of streets and misspelling makes it hard to describe all possible regular expressions to handle the problem. This results in difficulty to perform deterministic linking based on addresses.

5.3. Distance-based linkage

As the pairing of establishments using address data did not produce the expected results, the Netherlands proposed to use latitude and longitude information to combine data on tourist accommodation establishments from different sources.

Using this solution WPJ developed a tool for geolocation of address data from accommodation establishments and web scraping was developed. The solution was prepared in JavaScript language and uses HERE Maps API. HERE Maps are precise and up to date, so they are often used as car navigation. The variety of available functionalities in the application used, as well as extensive documentation and offered technical support allowed to build a universal tool being a part of the prototype created by WPJ. A very important function used in the developed solution is automatic parsing of address data to a common structure, which significantly reduces the time needed to prepare batch files for the used tool.

If the geographical coordinates are available for the data from the survey as well as from the accommodation portal, then the distance-based approach can be applied. Using geographical coordinates has an advantage over addresses. For some cases, the number of the building is not identified correctly. For instance, *Malinowa 21 m. 12* was recognized as *Malinowa 21* instead of *Malinowa 21/12* and the portal may derive slightly different coordinates. In the case of deterministic matching, no match would occur even if there was exactly one establishment in the survey frame with the address *Malinowa 21/12*.

The distance-based approach can be applied in the following way:

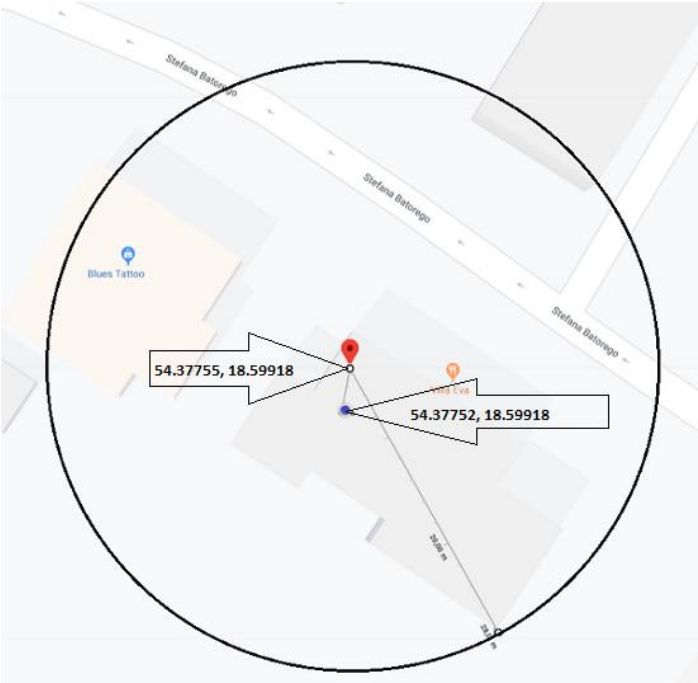
1. Derive geographical coordinates for all establishments in survey frame as well as in scraped data.
2. Calculate distance between all establishments in survey frame and all establishments in scraped data.
3. For each establishment in scraped data find all establishments in survey frame such that the distance between the establishment from scraped data and establishments in survey frame does not exceed a threshold
4. For each establishment in scraped data match the closest one found in the third step.

Several situations may happen since it is not exact matching:

1. True match is obtained – two establishments were paired correctly even if the distance was not zero.
2. False match is obtained – two establishments were paired due to low distance but in fact they are not the same.
3. True mismatch is obtained – two establishments were not paired due to high distance correctly,
4. False mismatch is obtained – two establishments were not paired due to high distance but in fact they were the same.

If the threshold is zero, then the first and the fourth situation may happen only. Increasing the threshold causes that there are more false matches as well as true mismatch – it is a trade-off. The true mismatches prevent underestimating of the number of establishments while false matches cause overestimating of the number of establishments. Setting the threshold plays a crucial role in this method. Figure 12 presents an example of correct matching.

Figure 12. Establishment matched correctly with different coordinates



The same establishment had slightly different coordinates in survey frame and in scraped data. Setting 20 m search radius enabled to find a match.

Quality of matching can be check by reviewing names and addresses of paired establishments and deriving the number of true matches and false matches.

5.4. Conclusions

The method presented above was tested on the example of data from Polish official statistics, i.e. a survey of accommodation establishments, data from the administrative register on non-categorized establishments and web scraping of the Hotels.com portal. The key to connecting these sets was the geographical coordinates with a critical distance of 100 m.

Then, such an automatically combined set of establishments was verified in terms of correct linkage of establishments, taking into account address, postal code and the name of the establishments.

The analysis showed that the initial assumptions for combining data by longitude and latitude are promising, but require further improvements. The distance-based approach correctly combines large accommodation establishments, i.e. hotels, hostels, etc. However, it is worth emphasising that the adopted 100 m threshold is too large to correctly link small accommodation establishments, i.e. apartments, guest rooms.

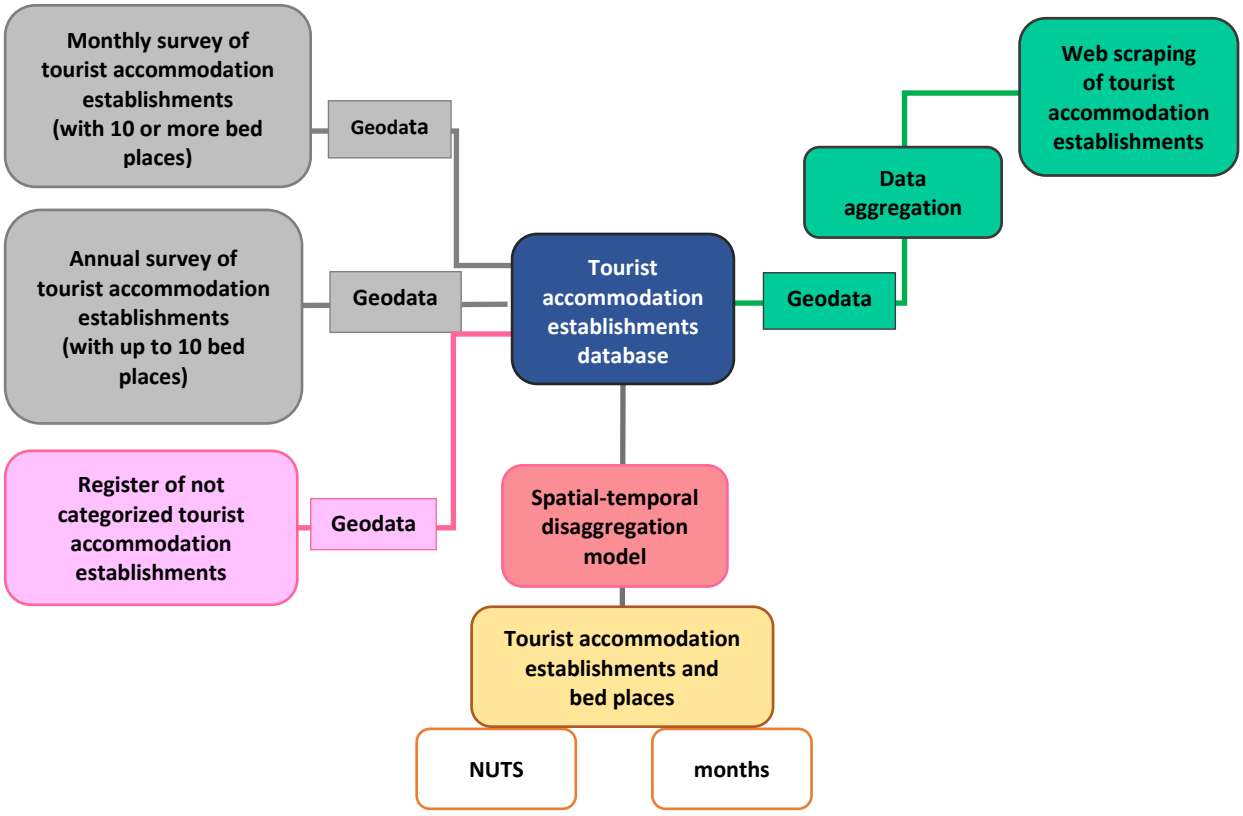
6. Spatial-temporal disaggregation of capacity of tourist accommodation establishments

Spatial-temporal disaggregation of capacity of tourist accommodation establishments may be carried out by integrating data from the survey and data from accommodation portals by data linkage and methods of data classification. The database of establishments created by data linkage of the statistical frames and scraped data (shortly *joint database*) is the key element of spatial–temporal disaggregation. After linkage there are accommodation establishments which are not covered by the frame. Thus, there are some missing variables which are needed to describe the capacity of establishments, e.g.:

- type of accommodation with respect to statistical classification,
- number of bed places,
- months in which establishment is operating.

Figure 13 shows data sources which can be used to derive these variables (Poland case). For the establishments which are surveyed (boxes marked in grey colour) or contained in official registers (pink colour) desired variables can be taken directly from the databases. For the establishments which are not surveyed but are present in scraped data (green colour) the variables can be obtained indirectly and imprecisely.

Figure 13. Capacity of tourist accommodation establishments – data processing



Accommodation type is presented in the portal but it does not necessarily matches the accommodation type in the statistical frame. A set of possible accommodation types in the portal usually overlaps only partially accommodation types used in official statistics. Moreover, accommodation establishment may be labelled *hotel* in the portal but in fact it is not a hotel according to law requirements.

It is usually not possible to derive a total number of bed places directly from the portal by a simple query. The reason is threefold:

- only some of the rooms are offered on the portal,
- some of the rooms offered on the portal are already booked,
- there is a limit on the number of adults in a query.

Hence, the number of bed places must be estimated in an indirect way. An auxiliary information about bed places from the portal can be obtained by setting several queries to a portal with different number of adults starting from one and finishing on the upper limit. It enables to derive the maximum available number of bed places via the portal including the upper limit.

Months in which accommodation establishment is operating can be derived from the presence of accommodation offer in a given month. If there is an offer, then the accommodation establishment operates in a given month. The reverse statement does not have to hold.

The capacity and the type of tourist accommodation establishment are obtained by web scraping with uncertainty and possible misclassification. These data can be verified and cleared based on additional information available in offers on the portal through model-based classification. Fortunately, scraped data contain some auxiliary variables which can be utilized for modelling:

- type of accommodation with respect to portal classification,
- price,
- star rating,
- guest review rating,
- total guest reviews.

Table 6 presents an example of the joint database. Data from the survey and from the portal are marked with orange and blue colour, respectively.

Table 6. Joint database of scraped data and survey frame

Hotel name	Bed places	Accommodation type from survey	Accommodation type from portal	Price	Star rating	Guest review rating	Guest review total
GEM Hotel	158	Hotel					
Dziki Potok	43	Guest house	Hotel	173.00	3	8.2	48
Lothus Hotel	175	Hotel	Hotel	233.00	3	7.6	106
Absynt Apartments Old Town Deluxe	60	Guest house	Apartment	150.00	4	7.6	118
WawaBed Bed & Breakfast			Bed and breakfast	289.00	4	9.2	13

The joint database contains a subset of establishments which is common for scraped data and statistical data. For these, there is information about the true accommodation type used in official

statistics as well as accommodation type used in the portal as well as other variables, such as geolocation, statistics on prices, etc. It should be noted that some establishments may be labelled as hotels in the portal but the survey revealed that part of them are hotels while others are guest houses. During assigning the accommodation type of the establishments not covered by the survey the difficulty of labelling the establishment as hotel or guest house occurs. An additional analysis of prices may reveal that on average hotels in a given location are priced higher than guest houses or hotels are reviewed more often than guest houses. Such analysis can provide decision rules of assigning accommodation type based on several variables. That common set of establishments can be used as a learning set in the classification model. One of the well-known classification method is a classification tree. Details of the method are provided in e.g. Breiman et al. (1984)[6]. Thanks to the classification model, the type of accommodation used in official statistics can be assigned to all establishments that are only available in web scraped data. A similar procedure can be applied for assigning the number of bed places or months when the accommodation establishments is operating.

Finally, using the classification tree, information about all required variables is available for all of the establishments in the joint database. Since there is geolocation data on address, NUTS, longitude and latitude coordinates, the data can be aggregated on any spatial level.

7. Combining data for expenditures estimation in country breakdown

7.1. General description of the methods

In almost every EU's country National Statistical Institutes conduct a household sample survey on participation of residents in trips. Due to the level of detail of a questionnaire or small subsample size for some destination countries, the research have large variability in expenditure as well as the problem of non-response related to expenditure by expenditure category with respect to some destination countries. If a country domain is not well represented then there may be no representative in a sample for some strata and the results, e.g. expenditures for business trips, cannot be estimated. In general, the following situations may occur:

1) There are some trips to the given country and some strata are not covered but there are historical data pertaining to these strata.
2) There are no trips to the given country in a given period but there are historical data.
3) New country appeared – there are no historical data.

It may be expected that the cases 1)-2) occurred just due to the insufficient sample size but in consequence there is no result for some domains, e.g. business purpose of the trip. In the third case probably very few trips are collected in a given period, hence the results for that country domain also may be of a low precision.

Within a framework of Big Data I project, a methodology for estimating the number of trips in air traffic was developed in the case of Poland. To this end the following data sources were used:

- data on flights obtained from 29 airports: 15 Polish airports and 14 hub airports (selected according to traffic intensity from Polish airports) were collected day-by-day with the web scraping tool. It consisted of origin and destination airport, IATA and ICAO codes, type of aircraft, date of arrival and departure.

[6] Breiman L., Friedman J. H., Olshen R. A., and Stone, C. J. (1984) Classification and Regression Trees. Wadsworth.

- list of over 10 thousand airports covering airport name, city, country, IATA, ICAO and FAA codes
- data on technical information on aircrafts covering type of aircraft, airline, ID, total number of seats and seats in each class.
- administrative data from Civil Aviation Authority of Poland on the number of passengers from Polish airports in domestic and foreign traffic. Foreign traffic data covered only direct flights from Poland which gives much less countries than from the sample survey.
- crucial statistics from the sample survey (share of Polish residents in total international air traffic, share of Polish residents using hub airports with respect to scraped hub airports).

Application of this method resulted with non-zero estimates for some countries which were not present in the sample survey. Hence, in consequence, there is a need to estimate expenditures for these new countries as a continuation of Big Data I project.

In this Chapter the methodology of estimating expenditures for trips to country domains which are not well-represented in a sample survey is proposed. The following assumptions are made:

- Number of trips to the given country is already obtained,
- Categorical variables, e.g. purpose of trip, means of transportation, type of accommodation, are given from historical data.

In this report general description of methodology is presented for the case when no additional source of information is available. The next report will cover the case when additional source of information is available for the country domains such as flight tickets or accommodation prices statistics.

In the first step the so-called learning dataset must be prepared. It is a set of historical data containing data for trips including descriptive variables such as destination country, purpose of trip, type of accommodation as well as expenditures by expenditure category. There must be no missing values in expenditures variables. In the second step it may be decided whether average expenditures per night are estimated or total expenditures are estimated. In the first case, in the learning set the average expenditure per night must be put instead of total expenditures. In the latter case, the variable: length of stay, must be included in the learning set while expenditures are expressed in totals.

When the learning set is prepared an imputation method is run.

The following methods may be used:

- Regression Trees (RT)[7],
- Support Vector Machine (SVM)[8],
- Random Forest (RF)[9],
- Predictive Mean Matching (PMM)[10],
- Bayesian Linear Regression (BLR)[11],
- Random Indicator for Nonignorable Data (RI)[12].

[7] Breiman L., Friedman J. H., Olshen R. A., and Stone, C. J. (1984) Classification and Regression Trees. Wadsworth.
[8] *Cristianini, Nello; Shawe-Taylor, John (2000). An Introduction to Support Vector Machines and other kernel-based learning methods. Cambridge University Press.*
[9] Ho TK (1998). The Random Subspace Method for Constructing Decision Forests. IEEE Transactions on Pattern Analysis and Machine Intelligence. 20 (8): 832–844.
[10] Rubin, Donald B. (30 June 1986). Statistical Matching Using File Concatenation with Adjusted Weights and Multiple Imputations. Journal of Business & Economic Statistics. 4 (1): 87–94.
[11] Box, G. E. P.; Tiao, G. C. (1973). Bayesian Inference in Statistical Analysis. Wiley.
[12] Jolani, S. (2012). Dual Imputation Strategies for Analyzing Incomplete Data. Dissertation. University of Utrecht.

These methods are available in *mice*, *e1071* and *rpart* packages in R.

The aforementioned imputation methods are based on some parameters and formulas or decision rules, which are derived from the learning set. Thus, these rules or formulas can be applied to impute data for the new dataset. That set is called the testing set because these data were not used to learn from. In fact, the true value of a given method can be examined on testing set rather than on learning set.

A comparison of the imputation methods can be made with the following accuracy measures: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE).

These measures can be applied to the learning and the testing set. A comparison of accuracy measures on both dataset may reveal if a given model is over-fitted to the learning set. It can be recognised by good performance on the learning set and bad performance on the training test at the same time.

Imputation by mean can be used as a reference point. In such case all missing values are imputed with mean calculated for all non-missing values in the database. In general, imputation methods based on modelling and additional sources of information should be better than imputation with mean. If accuracy of the imputation method is similar to the mean-based imputation, then that method is irrelevant or in the dataset there is a lot of randomness and lack of coherence.

7.2. Preliminary results

The objective of this research is to compare several statistical methods when dealing with missing values and estimating expenses for not well represented country domains. For this purpose five variables describing expenses were selected:

- expenses on accommodation,
- expenses on restaurants & cafés,
- expenses on transport,
- expenses on commodities,
- other expenses.

As explanatory variables, eleven categorical variables were used, including:

- destination country,
- purpose of trip,
- means of transportation,
- type of accommodation,
- accommodation booking method,
- nights spent,
- if the trip is organised or not,
- year and quarter, etc.

All of these explanatory variables had no missing values.

The preliminary training data set contained data on trips for the period 2016-2018. For each expenses variable, preliminary training data were trimmed to trips without missing values for the dependent

variable. Data for the first quarter of 2019 served as a testing set. Table 7 presents the size of learning and training sets for each variable.

Table 7. Size of datasets

Variable	Set	Size
Accommodation	Learning	10 491
	Testing	1 550
Restaurants & cafés	Learning	16 209
	Testing	774
Transport	Learning	17 123
	Testing	1 209
Commodities	Learning	17 565
	Testing	503
Other	Learning	7 993
	Testing	171

The datasets were loaded into R Studio. All of the compared methods are implemented in R in packages: *rpart*, *e1071* and *mice*. The methods were run with the same sets of explanatory variables and with default parameters. Imputation by mean was used as a reference point.

Table 8. Accuracy of mean-based imputation

Variable	Set	MAE	MAPE	RMSE
Accommodation	Learning	123.6	2.49	208.1
	Testing	129.6	2.73	209.8
Restaurants & café	Learning	66.2	1.64	132.2
	Testing	70.3	1.80	122.8
Transport	Learning	625.2	3.59	1 118.4
	Testing	733.3	2.47	1 347.2
Commodities	Learning	70.9	3.87	177.4
	Testing	68.3	4.50	153.8
Other	Learning	56.6	3.53	129.0
	Testing	68.4	2.57	183.8

Figures below present a comparison of methods with respect to MAPE. It may be noticed that expenses for accommodation and transport can be imputed quite well since MAPE for the imputation methods is lower than for the imputation by mean. The expenses for commodities and other expenses are hard to impute. Most of the imputation methods had nearly the same MAPE as mean-based imputation. Certainly, Support Vector Machine was a leader in every case and managed to do a better imputation of the expenses for commodities and other expenses, while the other methods made no value added.

Figure 14. Comparison of methods - MAPE on learning set

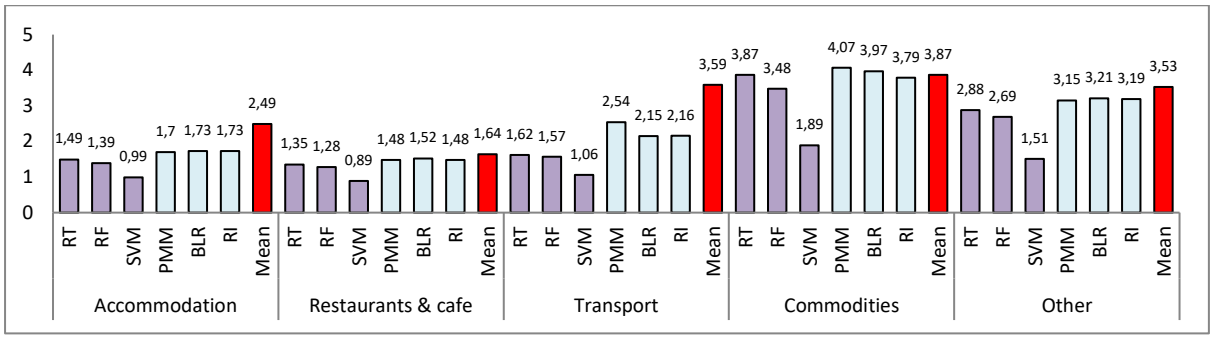
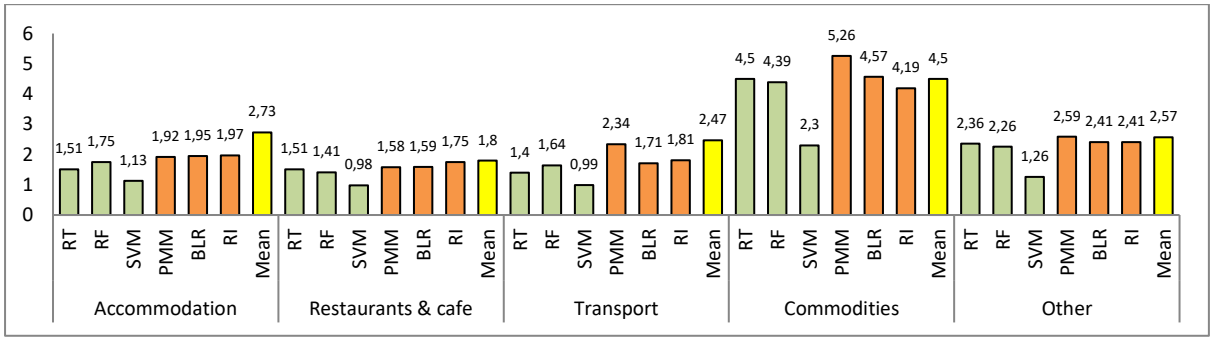


Figure 15. Comparison of methods - MAPE on testing set



Referring to the Big Data I results, for the comparison, countries from Latin America in the third quarter of 2017 were chosen. Based on the sample survey, the total number of trips amounted to 9 622 whereas in the case of big data sources to 10 513. The model for expenses was built with all the aforementioned explanatory variables. Table 9 presents trips and expenses from the sample survey, trips obtained in Big Data I and expenses estimated by the model.

Table 9. Comparison of trips and expenditures estimates with and without using big data

Country	Trips		Expenditures (PLN)	
	Big Data I	Survey	Big Data II	Survey
Dominican Republic	4 046	2 370	14 236 817	7 492 817
Mexico	2 548	982	12 099 156	3 732 507
Cuba	1 705	4 803	8 893 901	20 509 290
Brazil	1 064	1 114	8 101 153	3 441 436
Argentina	362	148	3 558 144	518 745
Chile	134	-	835 708	-
Colombia	129	-	861 556	-
Panama	113	-	706 567	-
Curacao	67	-	-	-
Saint Maarten	60	-	-	-
Suriname	50	-	-	-
Venezuela	39	-	124 857	-
Jamaica	33	-	221 422	-

Table 9. Comparison of trips and expenditures estimates with and without using big data (cont.)

Country	Trips		Expenditures (PLN)	
	Big Data I	Survey	Big Data II	Survey
Martinique	33	-	167 113	-
Ecuador	30	-	298 157	-
Guadeloupe	26	-	221 579	-
Aruba	21	-	-	-
Caribbean Netherlands	21	-	-	-
Bahamas	12	-	63 412	-
Trinidad and Tobago	12	-	2	-
Puerto Rico	7	-	15 034	-
Barbados	1	-	-	-
Paraguay	-	205	-	659 486
Total	10 513	9 622	50 404 578	36 354 281

At this moment, models enable to predict values for trips to countries that occurred at least once in the sample survey in the period 2016-2018 and have no missing values. Total amount of expenses from the model and from the survey amounted to 50.4 million and 36.3 million, respectively. It takes into account different number of trips. If the model's expenses were scaled to the number of trips in the sample survey, the total expenses from the model would amount to 46.1 million. The next step is to take into account other data sources, such as scraped data on flight ticket and accommodation costs to replace some variables from the learning set. Finally, it will enable prediction of expenses for other countries based on updated information rather than historical past data.

1) Some of the categories of expenses (e.g. on accommodation) can be predicted with better accuracy than mean-based imputation.
2) Additional information may be essential to improve imputation accuracy to avoid prediction based on historical data solely.
3) Machine Learning methods made a better imputation (based on MAPE) than non-Machine Learning methods. Support Vector Machine turned out to the leader in all of the cases.

8. Conclusions

The work carried out so far in the project concerned the analysis of preliminary results of inventory of data sources and web scraping procedures as well as methods related to combining data from various sources and spatial-temporal disaggregation.

A detailed inventory of big data sources potentially to be used in tourism statistics has shown that only about half of them can be used for the implementation of the project objectives. Some of the sources still need to be agreed with the administrators in terms of availability, others need to be further identified as to their quality and usability.

Therefore, further work on identifying data sources should be carried out. The sources related to modern technologies, e.g. Smart City sources, which will allow to compile statistics for small areas, as

well as to provide data on tourist traffic related to specific events or the character of a given place, may be interesting.

As part of the project work, a universal tool equipped with graphic interface has been developed, which allows the web scraping process to be easily performed. It is worth emphasizing that this is a proprietary and innovative solution that can be used by people who do not have IT expertise. This is particularly important because web scraping processes usually require a lot of work and knowledge of programming tools and languages, especially if it concerns different portals in many countries. Thanks to this solution, works related to web scraping will be standardized and accelerated.

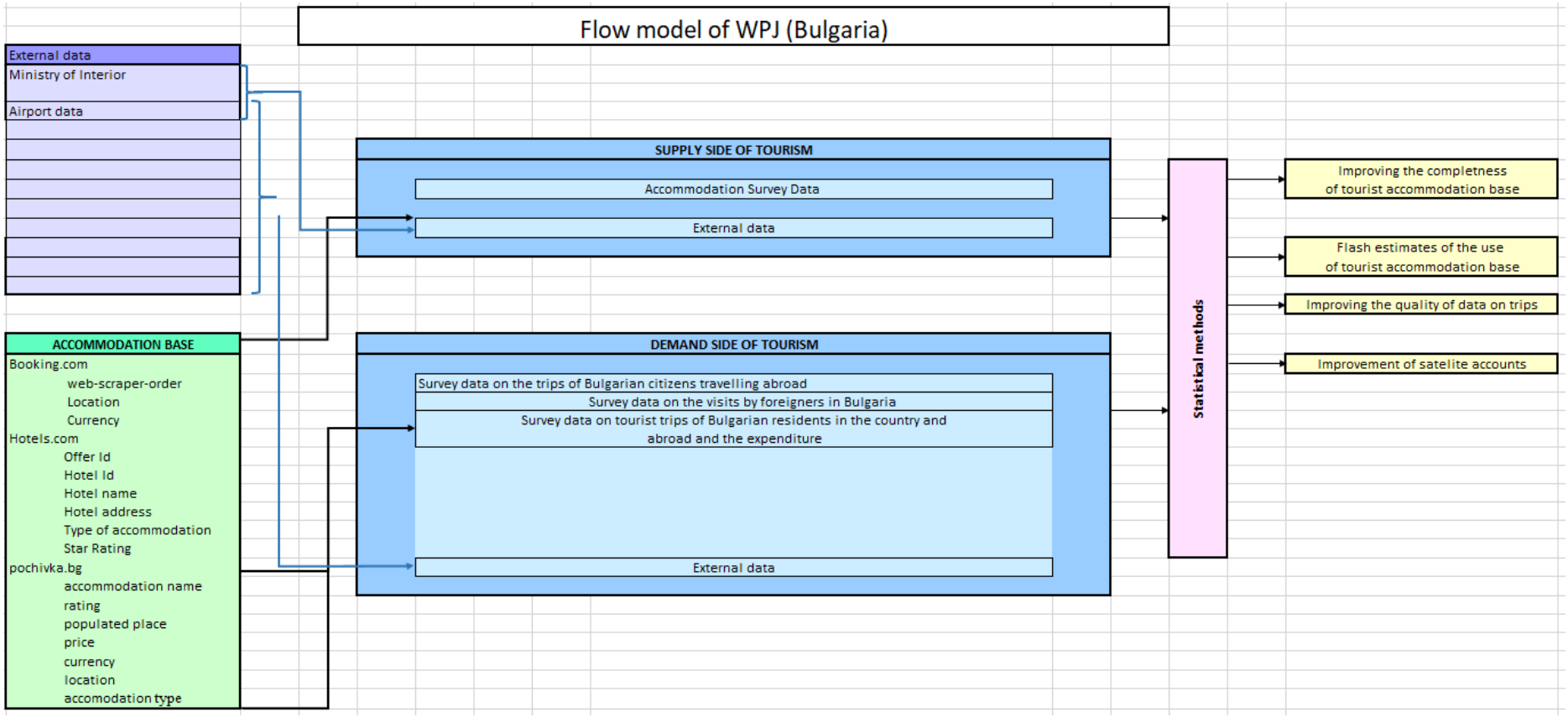
A very important stage of the work undertaken in the project is to develop appropriate methods for combining data from various sources. In this context, the use of geographical coordinates becomes extremely beneficial. This approach allows more flexible and precise matching of establishments.

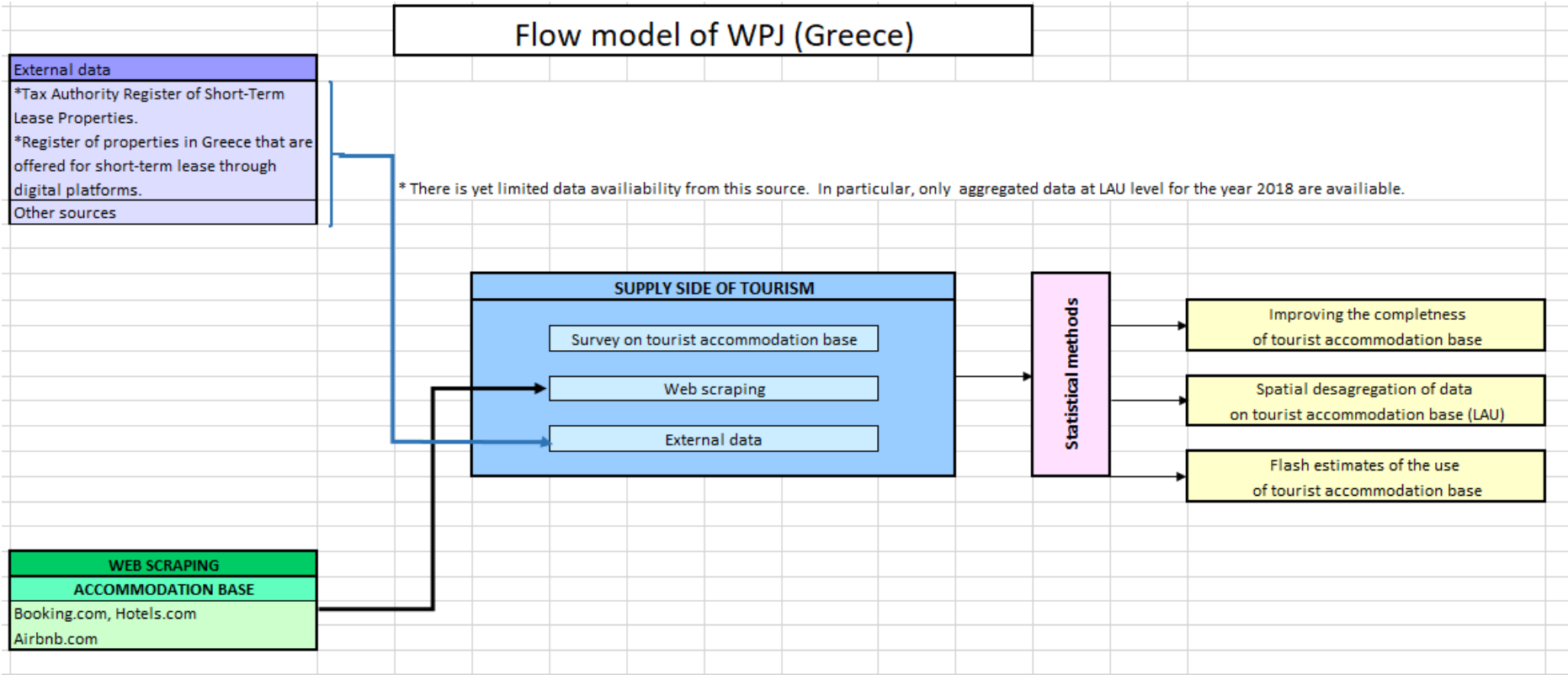
The Polish example shows that large tourist establishments have for the most part been correctly paired, while in the case of small establishments the quality of matching depends on the adopted value of the maximum difference resulting from the location of the tourist establishment based on geographical coordinates from different sources.

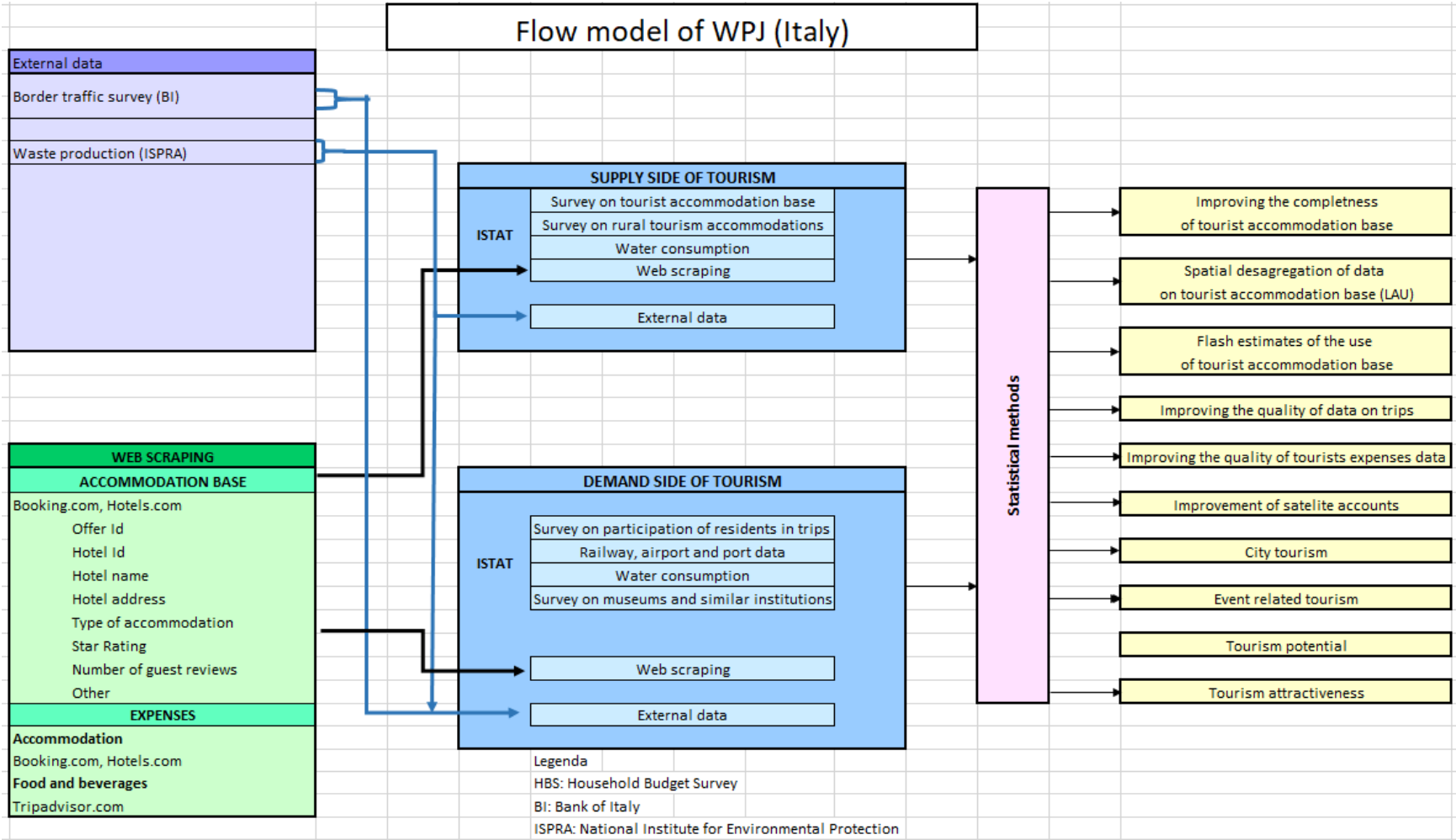
The project work also included the development of a conceptual model for estimating tourists expenditure based on Machine Learning methods, which are a continuation of the work undertaken in the Big Data I project. It should be noted that promising preliminary results were obtained. Further work carried out within the project will primarily concern testing and implementation of developed solutions in all countries participating in the project and the implementation of subsequent tasks, i.e. developing flash estimates in the field of tourism and using big data sources and developed methodology to improve the quality of data in various statistical areas.

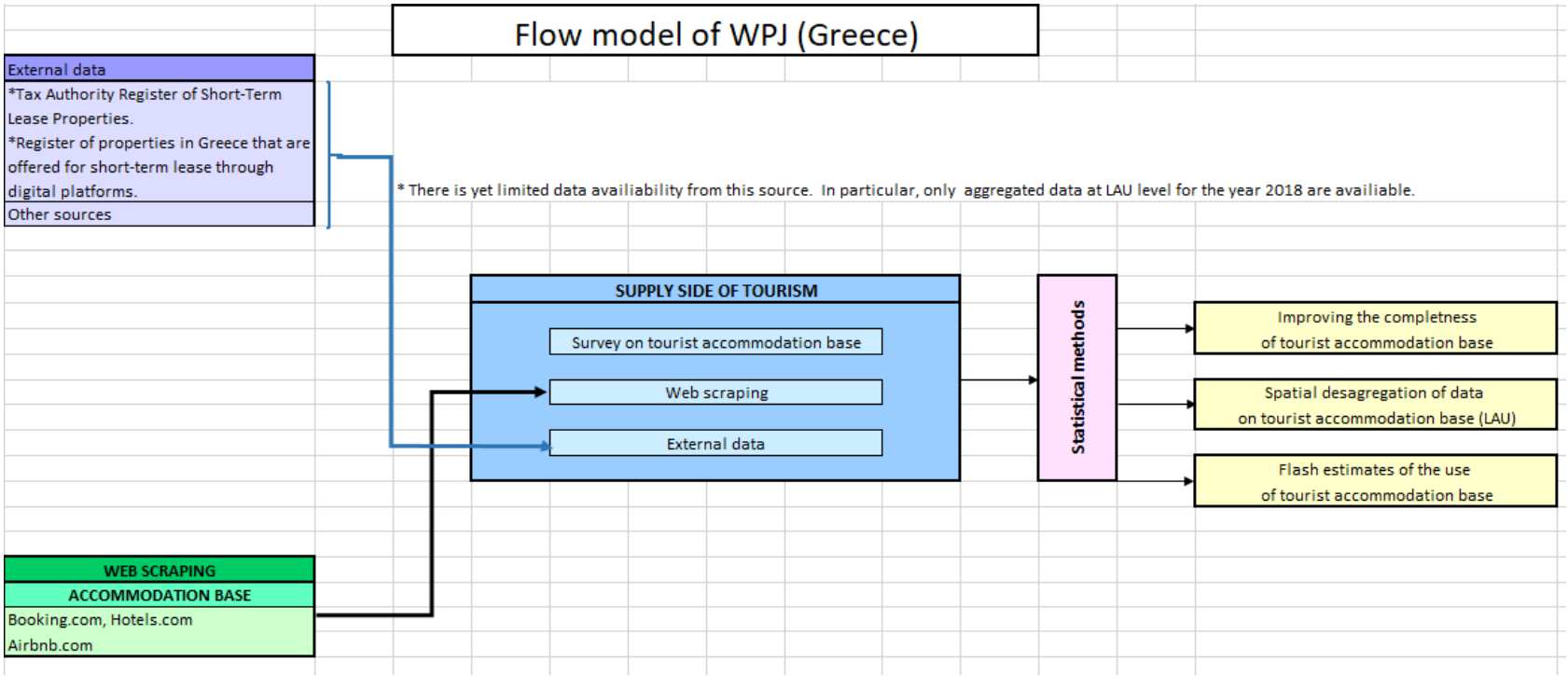
Annexes

Annex 1: Flow Models for WPJ partners




Flow model of WPJ (Greece)
External data
*Tax Authority Register of Short-Term Lease Properties.
*Register of properties in Greece that are offered for short-term lease through digital platforms.
Other sources
* There is yet limited data availiability from this source. In particular, only aggregated data at LAU level for the year 2018 are availiable.
SUPPLY SIDE OF TOURISM
Survey on tourist accommodation base
Web scraping
External data
Statistical methods
Improving the completness of tourist accommodation base
Spatial desagregation of data on tourist accommodation base (LAU)
Flash estimates of the use of tourist accommodation base
WEB SCRAPING
ACCOMMODATION BASE
Booking.com, Hotels.com
Airbnb.com


Flow model of WPJ (Italy)
External data
Border traffic survey (BI)
Waste production (ISPRA)
WEB SCRAPING
ACCOMMODATION BASE
Booking.com, Hotels.com
Offer Id
Hotel Id
Hotel name
Hotel address
Type of accommodation
Star Rating
Number of guest reviews
Other
EXPENSES
Accommodation
Booking.com, Hotels.com
Food and beverages
Tripadvisor.com
SUPPLY SIDE OF TOURISM
ISTAT
Survey on tourist accommodation base
Survey on rural tourism accommodations
Water consumption
Web scraping
External data
DEMAND SIDE OF TOURISM
ISTAT
Survey on participation of residents in trips
Railway, airport and port data
Water consumption
Survey on museums and similar institutions
Web scraping
External data
Statistical methods
Improving the completness of tourist accommodation base
Spatial desagregation of data on tourist accommodation base (LAU)
Flash estimates of the use of tourist accommodation base
Improving the quality of data on trips
Improving the quality of tourists expenses data
Improvement of satelite accounts
City tourism
Event related tourism
Tourism potential
Tourism attractiveness
Legenda
HBS: Household Budget Survey
BI: Bank of Italy
ISPRA: National Institute for Environmental Protection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* There is yet limited data availiability from this source. In particular, only aggregated data at LAU level for the year 2018 are availiable.
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ACCOMMODATION BASE
Booking.com, Hotels.com
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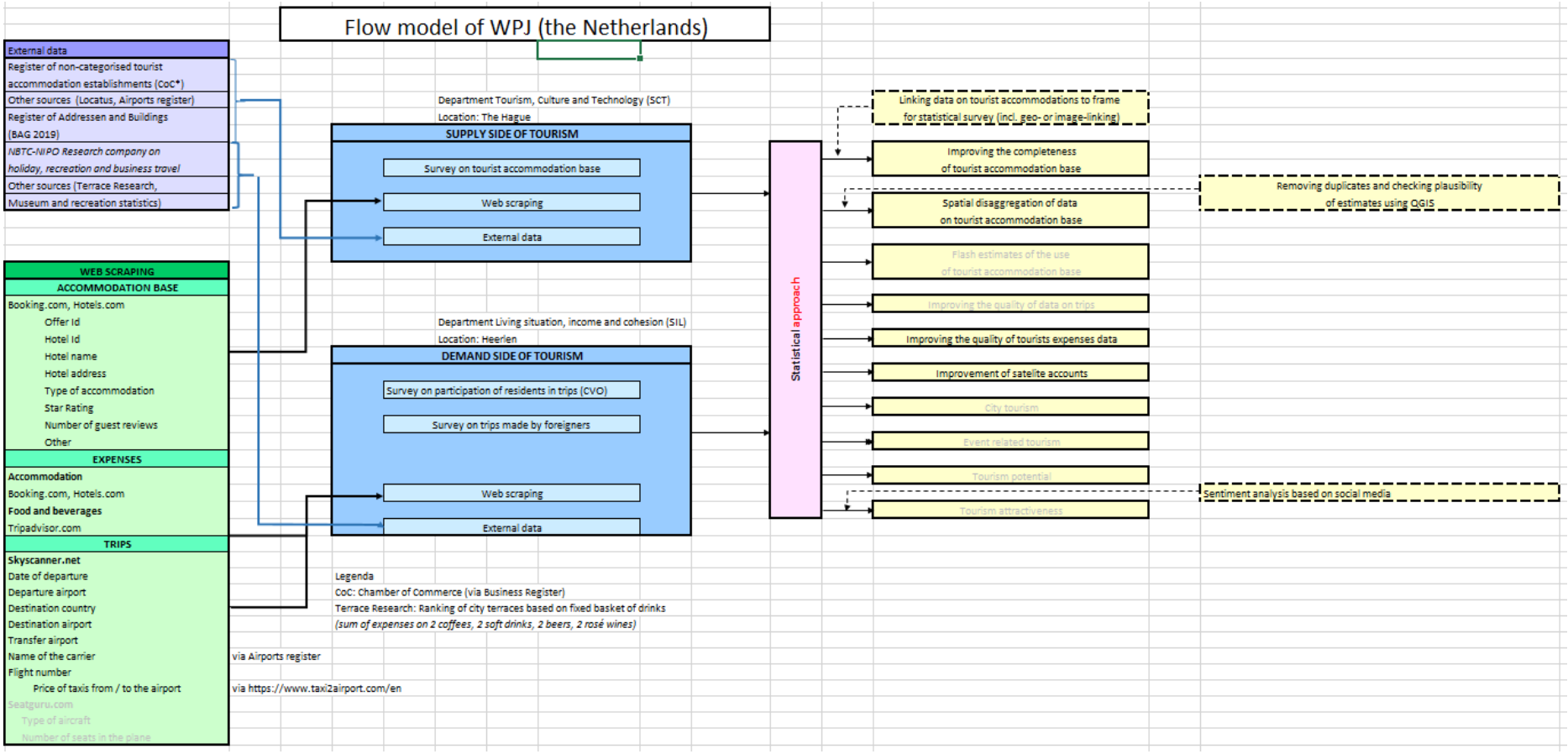
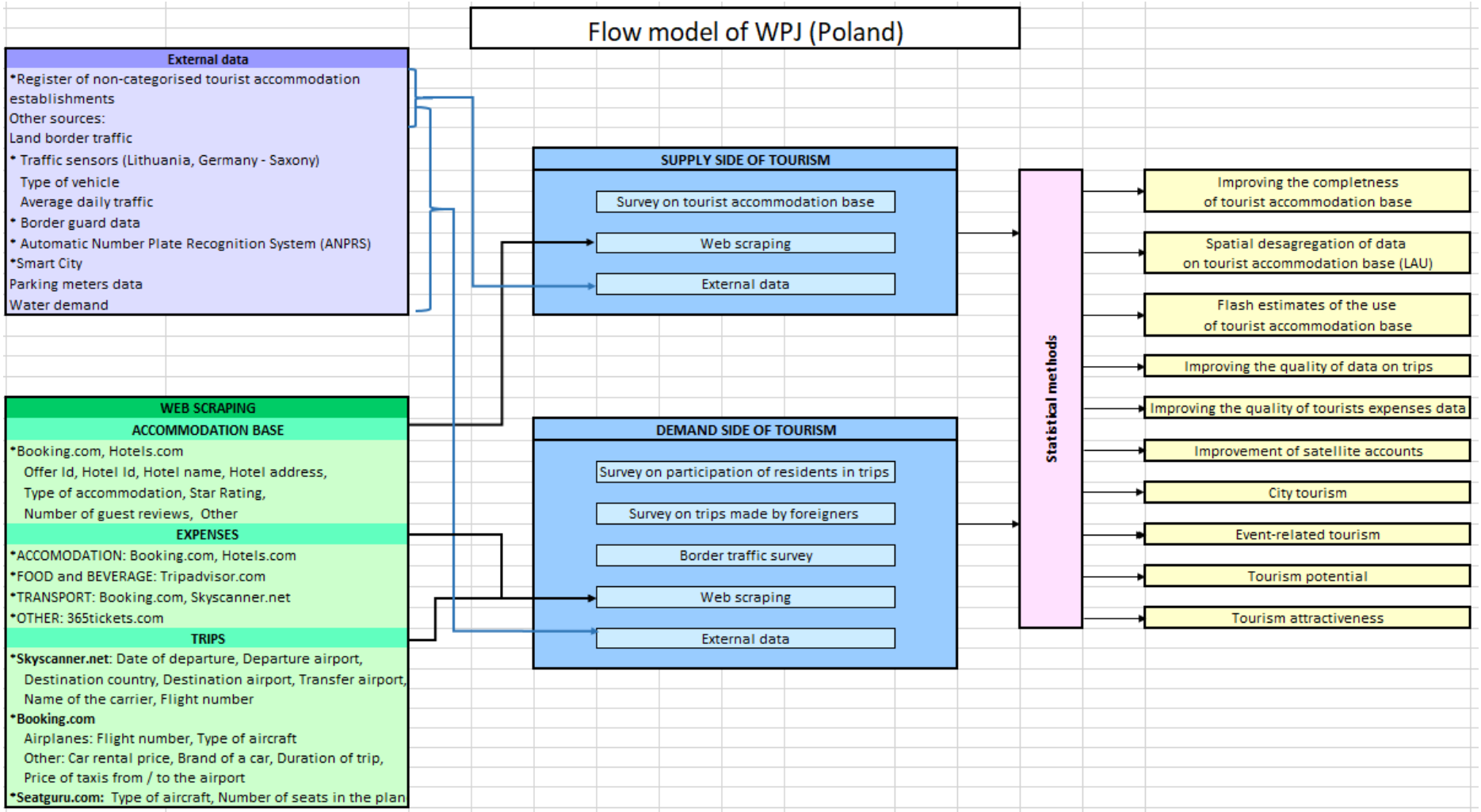

Flow model of WPJ (the Netherlands)
External data
Register of non-categorised tourist accommodation establishments (CoC*)
Other sources (Locatus, Airports register)
Register of Addressen and Buildings (BAG 2019)
NBTC-NIPO Research company on holiday, recreation and business travel
Other sources (Terrace Research, Museum and recreation statistics)
WEB SCRAPING
ACCOMMODATION BASE
Booking.com, Hotels.com
Offer Id
Hotel Id
Hotel name
Hotel address
Type of accommodation
Star Rating
Number of guest reviews
Other
EXPENSES
Accommodation
Booking.com, Hotels.com
Food and beverages
Tripadvisor.com
TRIPS
Skyscanner.net
Date of departure
Departure airport
Destination country
Destination airport
Transfer airport
Name of the carrier
Flight number
Price of taxis from / to the airport
Seatguru.com
Type of aircraft
Number of seats in the plane
Department Tourism, Culture and Technology (SCT)
Location: The Hague
SUPPLY SIDE OF TOURISM
Survey on tourist accommodation base
Web scraping
External data
Department Living situation, income and cohesion (SIL)
Location: Heerlen
DEMAND SIDE OF TOURISM
Survey on participation of residents in trips (CVO)
Survey on trips made by foreigners
Web scraping
External data
Legenda
CoC: Chamber of Commerce (via Business Register)
Terrace Research: Ranking of city terraces based on fixed basket of drinks
(sum of expenses on 2 coffees, 2 soft drinks, 2 beers, 2 rosé wines)
via Airports register
via https://www.taxi2airport.com/en
Statistical approach
Linking data on tourist accommodations to frame for statistical survey (incl. geo- or image-linking)
Improving the completeness of tourist accommodation base
Spatial disaggregation of data on tourist accommodation base
Flash estimates of the use of tourist accommodation base
Improving the quality of data on trips
Improving the quality of tourists expenses data
Improvement of satelite accounts
City tourism
Event related tourism
Tourism potential
Tourism attractiveness
Removing duplicates and checking plausibility of estimates using QGIS
Sentiment analysis based on social media


Flow model of WPJ (Poland)
External data
*Register of non-categorised tourist accommodation establishments
Other sources:
Land border traffic
* Traffic sensors (Lithuania, Germany - Saxony)
Type of vehicle
Average daily traffic
* Border guard data
* Automatic Number Plate Recognition System (ANPRS)
*Smart City
Parking meters data
Water demand
WEB SCRAPING
ACCOMMODATION BASE
*Booking.com, Hotels.com
Offer Id, Hotel Id, Hotel name, Hotel address,
Type of accommodation, Star Rating,
Number of guest reviews, Other
EXPENSES
*ACCOMODATION: Booking.com, Hotels.com
*FOOD and BEVERAGE: Tripadvisor.com
*TRANSPORT: Booking.com, Skyscanner.net
*OTHER: 365tickets.com
TRIPS
*Skyscanner.net: Date of departure, Departure airport,
Destination country, Destination airport, Transfer airport,
Name of the carrier, Flight number
*Booking.com
Airplanes: Flight number, Type of aircraft
Other: Car rental price, Brand of a car, Duration of trip,
Price of taxis from / to the airport
*Seatguru.com: Type of aircraft, Number of seats in the plan
SUPPLY SIDE OF TOURISM
Survey on tourist accommodation base
Web scraping
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DEMAND SIDE OF TOURISM
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Survey on trips made by foreigners
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City tourism
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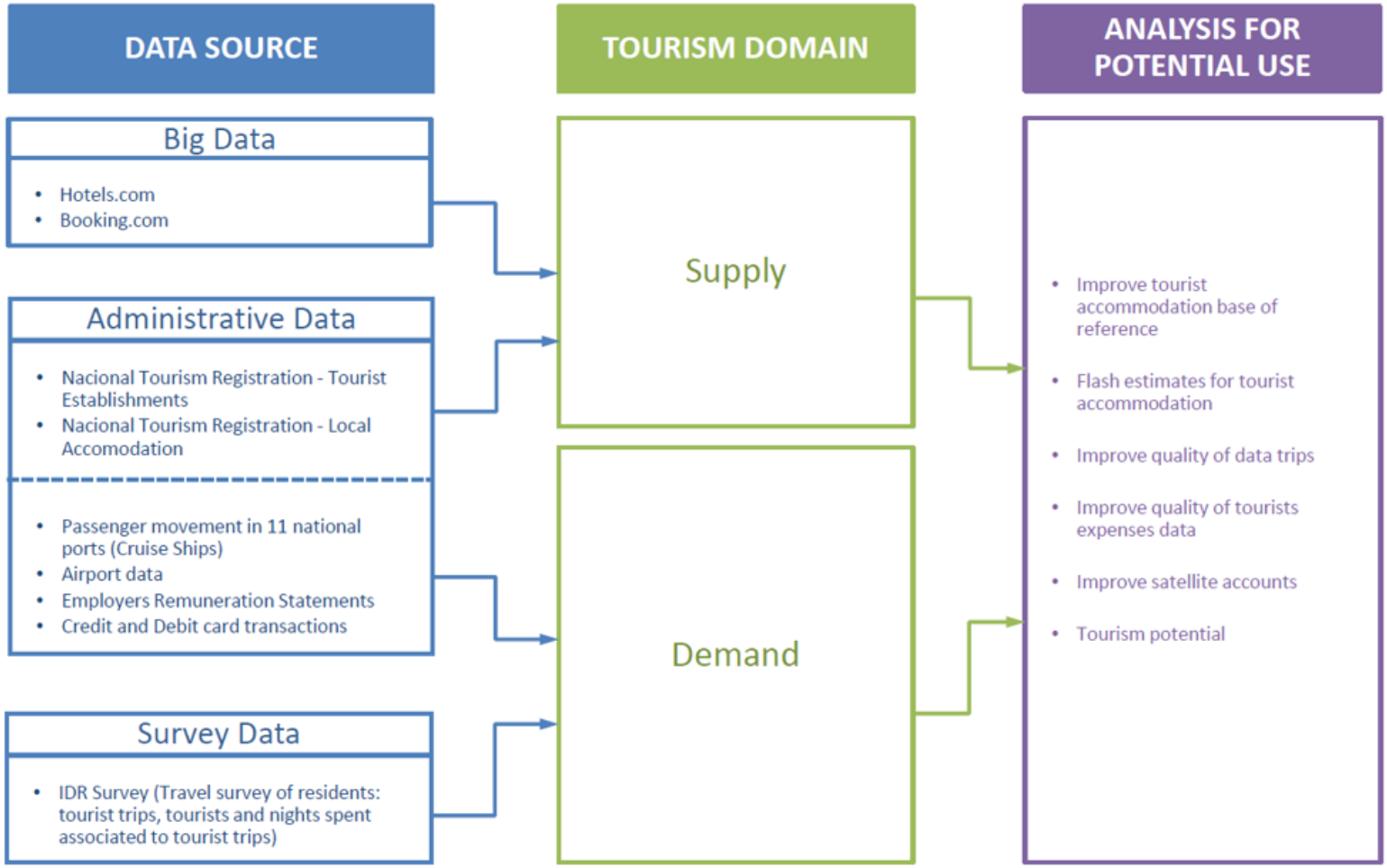
Tourism attractiveness



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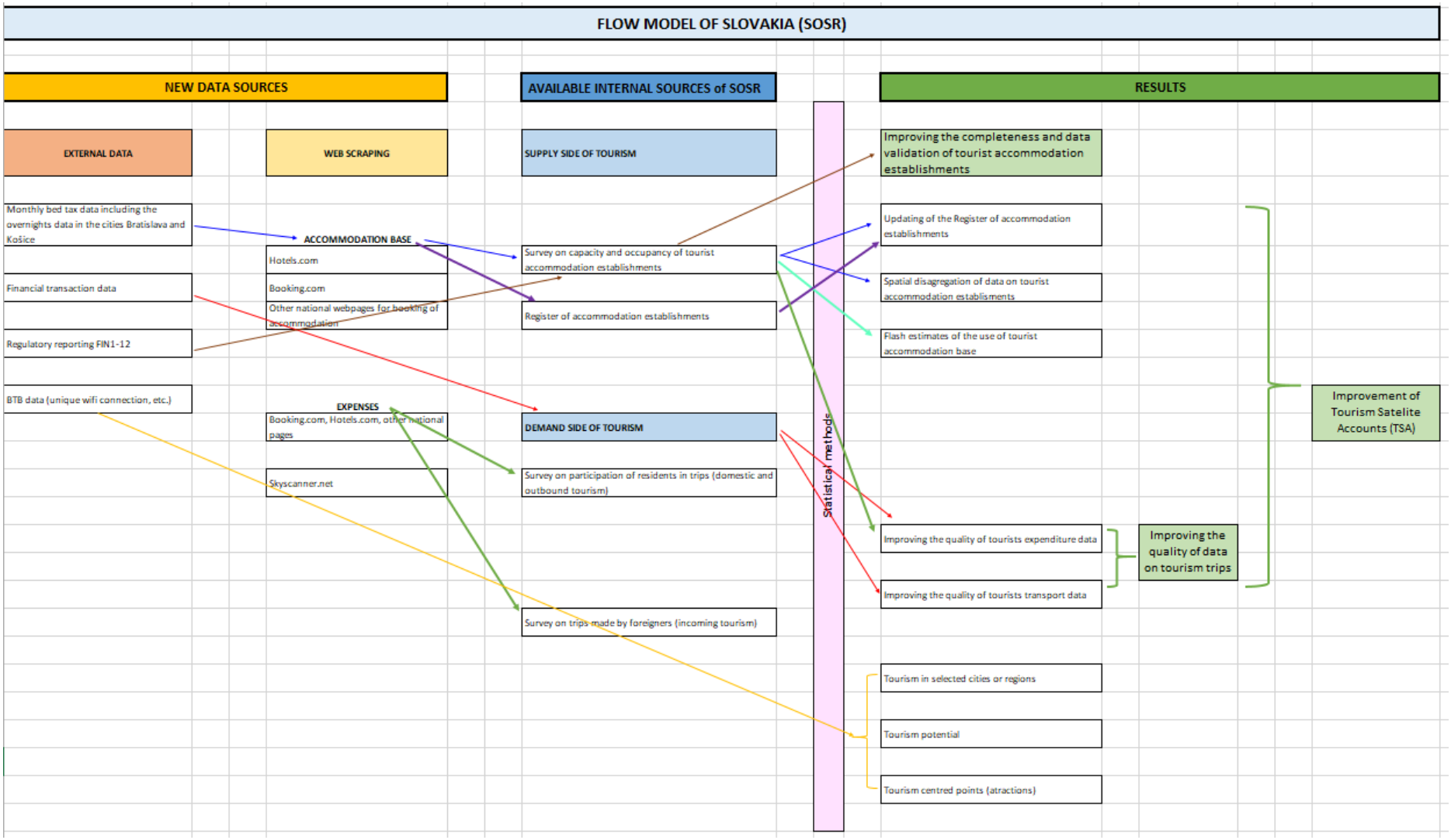
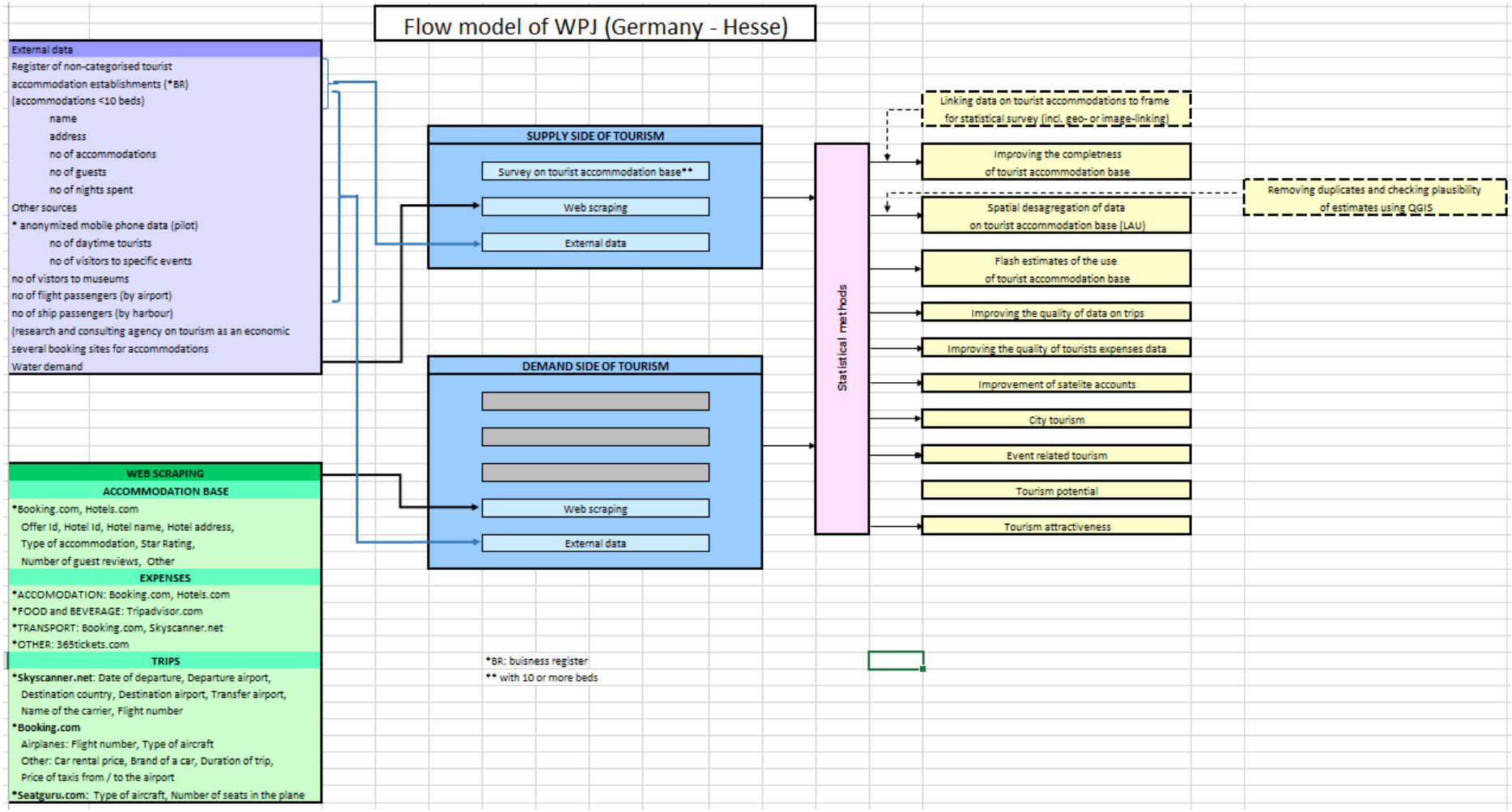
STATISTICS PORTUGAL

ESSnet BD II – WPJ – Portuguese workflow (v.2)


FLOW MODEL OF SLOVAKIA (SOSR)
NEW DATA SOURCES
AVAILABLE INTERNAL SOURCES of SOSR
RESULTS
EXTERNAL DATA
WEB SCRAPING
SUPPLY SIDE OF TOURISM
Monthly bed tax data including the overnights data in the cities Bratislava and Košice
ACCOMMODATION BASE
Hotels.com
Booking.com
Other national webpages for booking of accommodation
Financial transaction data
Regulatory reporting FIN1-12
BTB data (unique wifi connection, etc.)
EXPENSES
Booking.com, Hotels.com, other national pages
Skyscanner.net
Survey on capacity and occupancy of tourist accommodation establishments
Register of accommodation establishments
DEMAND SIDE OF TOURISM
Survey on participation of residents in trips (domestic and outbound tourism)
Survey on trips made by foreigners (incoming tourism)
Statistical methods
Improving the completeness and data validation of tourist accommodation establishments
Updating of the Register of accommodation establishments
Spatial disagregation of data on tourist accommodation establisments
Flash estimates of the use of tourist accommodation base
Improving the quality of tourists expenditure data
Improving the quality of tourists transport data
Tourism in selected cities or regions
Tourism potential
Tourism centred points (atractions)
Improving the quality of data on tourism trips
Improvement of Tourism Satelite Accounts (TSA)


Flow model of WPJ (Germany - Hesse)
External data
Register of non-categorised tourist
accommodation establishments (*BR)
(accommodations <10 beds)
name
address
no of accommodations
no of guests
no of nights spent
Other sources
* anonymized mobile phone data (pilot)
no of daytime tourists
no of visitors to specific events
no of vistors to museums
no of flight passengers (by airport)
no of ship passengers (by harbour)
(research and consulting agency on tourism as an economic
several booking sites for accommodations
Water demand
SUPPLY SIDE OF TOURISM
Survey on tourist accommodation base**
Web scraping
External data
DEMAND SIDE OF TOURISM
Web scraping
External data
WEB SCRAPING
ACCOMMODATION BASE
*Booking.com, Hotels.com
Offer Id, Hotel Id, Hotel name, Hotel address,
Type of accommodation, Star Rating,
Number of guest reviews, Other
EXPENSES
*ACCOMODATION: Booking.com, Hotels.com
*FOOD and BEVERAGE: Tripadvisor.com
*TRANSPORT: Booking.com, Skyscanner.net
*OTHER: 365tickets.com
TRIPS
*Skyscanner.net: Date of departure, Departure airport,
Destination country, Destination airport, Transfer airport,
Name of the carrier, Flight number
*Booking.com
Airplanes: Flight number, Type of aircraft
Other: Car rental price, Brand of a car, Duration of trip,
Price of taxis from / to the airport
*Seatguru.com: Type of aircraft, Number of seats in the plane
*BR: buisness register
** with 10 or more beds
Statistical methods
Linking data on tourist accommodations to frame
for statistical survey (incl. geo- or image-linking)
Improving the completness
of tourist accommodation base
Removing duplicates and checking plausibility
of estimates using QGIS
Spatial desagregation of data
on tourist accommodation base (LAU)
Flash estimates of the use
of tourist accommodation base
Improving the quality of data on trips
Improving the quality of tourists expenses data
Improvement of satelite accounts
City tourism
Event related tourism
Tourism potential
Tourism attractiveness

Annex 2: CBS taxonomy – Word file

CBS Nederlands toerisme
toerisme @ CBS taxonomie
Broader Terms
BT recreatie en cultuur
More specific terms
NT2 accommodatiegebruik
NT2 bestedingen toeristen
NT2 logiesaccommodaties
NT2 logiesvormen
NT2 reisgezelschap
NT2 toerismebeleid
NT2 toerismerekeningen
NT2 toeristen
NT2 toeristengebieden
NT2 vakanties van Nederlanders
NT2 zakelijk toerisme
Related terms
RT reisbranche
Non preferred terms
UF vreemdelingenverkeer

CBS English tourism
tourism @ CBS taxonomy
Broader Terms
BT recreation and culture
More specific terms
NT business travel
NT holidays by Dutch people
NT overnight accommodation
NT tour group
NT tourism accounts
NT tourism policy
NT tourist areas
NT tourist expenditure
NT tourists
NT types of accommodation
NT use of accommodation
Related terms
RT travel industry
Non preferred terms
UF foreigners traffic

CBS Nederlands gebruik van accommodaties
accommodatiegebruik @ CBS taxonomie
Broader Terms
BT toerisme
More specific terms
NT3 overnachtingen
NT3 verblijfsduur

CBS English use of accommodation
use of accommodation @ CBS taxonomy
Broader Terms
BT tourism
More specific terms
NT length of stay
NT overnight stays

CBS Nederlands bestedingen toeristen
bestedingen toeristen @ CBS taxonomie
Non preferred terms
UF toeristische bestedingen

CBS English tourist expenditure
tourist expenditure @ CBS taxonomy
Non preferred terms
UF tourist spending

Annex 3: R script to VisNetwork object – Word file

```
# Flowchart on Methodological manual for tourism statistics ---------------
# R version 3.6.1 (2019-07-05)
# RStudio version 1.2.1335
# load packages ------------------------------------------------------------
library(dplyr)
library(visNetwork)
library(networkD3)
# create data --------------------------------------------------------------
my_nodes <-
        structure(list(id = c("Travel", "Non - Tourism", "Tourism", "Same-day trip",
"Overnight trip", "Inbound tourism", "Outbound tourism", "Domestic tourism",
"Hosehold surv", "Tourism stat", "Accommodation stat", "Border surv", "Other stat"),
label = c("Travel", "Non - Tourism", "Tourism","Same-day trip", "Overnight trip", "Inbound
tourism", "Outbound tourism", "Domestic tourism", "Hosehold surv", "Tourism stat",
"Accommodation stat", "Border surv", "Other stat"), group = c("concept", "concept",
"concept", "concept", "concept", "concept", "concept", "source", "source", "source",
"source", "source", "source")), row.names = c(NA, -13L), class = "data.frame")

my_edges <-
        structure(list(from = c("Travel", "Travel", "Tourism", "Tourism", "Same-day trip",
"Same-day trip", "Same-day trip", "Overnight trip", "Overnight trip", "Overnight trip",
"Tourism stat", "Hosehold surv", "Border surv", "Accommodation stat", "Border surv", "Border
surv", "Hosehold surv", "Hosehold surv", "Accommodation stat", "Accommodation stat"), to =
c("Non - Tourism", "Tourism", "Same-day trip", "Overnight trip", "Inbound tourism", "Outbound
tourism", "Domestic tourism", "Inbound tourism", "Outbound tourism", "Domestic tourism",
"Tourism", "Same-day trip", "Same-day trip", "Overnight trip", "Inbound tourism", "Outbound
tourism", "Outbound tourism", "Domestic tourism", "Inbound tourism", "Domestic tourism"),
width = c(3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1), ...4 = c("Travel",
"Non - Tourism", "Tourism", "Same-day trip", "Overnight trip", "Inbound tourism", "Outbound
tourism", "Domestic tourism", NA, "Hosehold surv", "Tourism stat", "Accommodation stat",
"Border surv", "Other stat", NA, NA, NA, NA, NA, NA)), row.names = c(NA, -20L), class =
"data.frame")

# create VisNetwork object -------------------------------------------------
visNetTest <-
        visNetwork(my_nodes, my_edges) %>%
        visIgraphLayout() %>%
        visNodes(shape = "dot",color = list(background = "#0085AF",border =
"#013848",highlight = "#FF8000"),shadow = list(enabled = TRUE, size = 10), labelHighlightBold
= T) %>%
        visEdges(shadow = FALSE, color = list(color = "#0085AF", highlight = "#C62F4B"),
                 arrows = 'to') %>%
        visOptions(highlightNearest = list(enabled = T, degree = 1, hover = T), selectedBy
= "group") %>%
        visLayout(randomSeed = 11, improvedLayout = T, hierarchical = T) %>%
        visHierarchicalLayout(direction = "UD", sortMethod = "directed", nodeSpacing = 400,
levelSeparation = 200) %>%
        visClusteringByGroup(groups = c("concept", "source")) %>%
        visLegend()

visNetTest # see result on RStudio viewer
visNetTest %>% saveNetwork(file = 'visNetTest.html') # export to html
# ------------------------------------------------------------------------
```

Annex 4: R script for redirecting networkd3 node click to a dynamic url based on a value – Word file

```
# Redirecting networkd3 node click to a dynamic url based on a val --------
# For more information on this example please see following links:
# https://christophergandrud.github.io/networkD3/
#       https://stackoverflow.com/questions/49640151/redirecting-networkd3-node-click-to-a-
dynamic-url-based-on-a-value

# load packages ------------------------------------------------------------
library(networkD3)
library(magrittr)

# load data ----------------------------------------------------------------
data(MisLinks)
data(MisNodes)

# create object ------------------------------------------------------------
fn <- forceNetwork(
        Links = MisLinks, Nodes = MisNodes, Source = "source",
        Target = "target", Value = "value", NodeID = "name",
        Group = "group", opacity = 0.4, zoom = TRUE)
fn$x$nodes$hyperlink <- paste0(
        'http://en.wikipedia.org/wiki/Special:Search?search=',
        MisNodes$name)

fn$x$options$clickAction = 'window.open(d.hyperlink)' # define action on click

fn %>% saveNetwork(file = 'Network_LesMiserables.html') # export to html
# ------------------------------------------------------------------------
```

References

Box, G. E. P.; Tiao, G. C. (1973). Bayesian Inference in Statistical Analysis. Wiley.

Breiman L., Friedman J. H., Olshen R. A., and Stone, C. J. (1984) Classification and Regression Trees. Wadsworth.

Cristianini, Nello; Shawe-Taylor, John (2000). An Introduction to Support Vector Machines and other kernel-based learning methods. Cambridge University Press.

Ho TK (1998). The Random Subspace Method for Constructing Decision Forests. IEEE Transactions on Pattern Analysis and Machine Intelligence. 20 (8): 832–844.

Jolani, S. (2012). Dual Imputation Strategies for Analyzing Incomplete Data. Dissertation. University of Utrecht.

Rubin, Donald B. (30 June 1986). Statistical Matching Using File Concatenation with Adjusted Weights and Multiple Imputations. Journal of Business & Economic Statistics. 4 (1): 87–94.

https://ec.europa.eu/eurostat/cros/system/files/s-dwh-m_4.2_methodology_data_linkage_v2.pdf [Accessed 24.08.2019]

https://webgate.ec.europa.eu/fpfis/wikis/display/EstatBigData/Big+Data+Typification [Accessed 24.09.2019]

https://protege.stanford.edu/ [Accessed 24.09.2019]

https://christophergandrud.github.io/networkD3/ [Accessed 24.09.2019]

https://datastorm-open.github.io/visNetwork/ [Accessed 24.09.2019]

Glossary

API (Application Programming Interface) is a way, understood as a strictly defined set of rules and their descriptions, in which computer programs communicate with each other.

XML (Extensible Markup Language) is a universal markup language designed to represent different data in a structured way. It is platform-independent, which facilitates the exchange of documents between different systems.

HTML (HyperText Markup Language) is a hypertext markup language that allows one to describe the structure of information contained within a web page, giving meaning to individual fragments of text – forming hyperlinks, paragraphs, headers and lists - and embedding file objects in the text

XLSX (Office Open XML) is a zipped, XML-based file format developed by Microsoft for representing spreadsheets, charts, presentations and word processing documents.

Web scraping (also known as Screen Scraping, Web Data Extraction, Web Harvesting, etc.) is a technique used to extract large amounts of data from websites.